

Local-Unitary Rigidity of Stabilizer AME States and Transversal Clifford Groups of MDS–CSS Codes

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Abstract

Let $q = p^e$. For every $m \geq 2$, each product-unitary intertwiner between two stabilizer $\text{AME}(2m, q)$ states is Clifford factor by factor. On any $m + 1$ parties, the supported stabilizer labels form a q^2 -element group and project bijectively onto the complete local Weyl basis at every party in the marginal. The reduced operator therefore determines its local Weyl axes intrinsically. Consequently, every tensor-product conversion between the associated $[[2m - 1, 1, m]]_q$ stabilizer quantum-MDS encoders is Clifford on every physical factor and on the logical factor. Discreteness of these symmetry groups needs neither maximal entanglement nor stabilizer structure: every pure state whose two-party marginals are all maximally mixed, at any local dimension, has finite product-unitary symmetry group modulo phase, and the defect of a product unitary grows quadratically away from that group with an explicit ratio constant independent of the number of parties. The certified set is described in generator coordinates: its ℓ^1 radius is governed by the state's uniformity order. For a stabilizer state the exact product-Clifford symmetries are not merely isolated: any product Clifford outside the symmetry group has defect at least $(2 - 2p^{-1/2})^{1/2}$, a constant of the characteristic alone. Quantitative Clifford rounding and that separation remove compactness from the approximate-symmetry decomposition and give defect-only thresholds. Viewing each party as the logical leg of an $[[2m - 1, 1, m]]_q$ code, a quantitative three-region cleaning argument rounds every local factor at constant error and gives a global threshold of order $m^{-1/2}$ for fixed local dimension, conditional on AME existence. This bypasses the exponentially faint minimum-support marginals. Modulo party relabelling and local trace-symplectic $\mathbb{F}_p$ -linear changes of frame, the transition atlas obtained by pushing Weyl labels between parties on minimum supports is a complete structured LU invariant.

For equal-phase CSS states of linear $[2m, m, m + 1]_q$ MDS codes over an odd prime field, the exact fixed-party projective transversal logical group is $\mathbb{F}_q^2 \rtimes \text{SL}_2(q)$ precisely when C is diagonally isodual, and $\mathbb{F}_q^2 \rtimes T$ otherwise, where T is the split torus; generalized Reed–Solomon codes lie on the first branch.

For $m = 3$, a degree-eight quotient z classifies projective and monomial-code equivalence on an explicit non-GRS pencil of six-point arcs over odd fields; over odd prime fields it also classifies local-Clifford and local-unitary equivalence.

1 Introduction

An absolutely maximally entangled state has maximally mixed reductions on every subsystem no larger than half the parties. Fix a prime power $q = p^e$ and the standard additive q -dimensional Weyl system. We ask whether a stabilizer $\text{AME}(2m, q)$ state retains its local Weyl frames under arbitrary single-party changes of basis.

The assertion that local-unitary (LU) equivalence implies local-Clifford (LC) equivalence fails for general stabilizer states [JCWY10]. Maximal entanglement supplies the missing rigidity.

Throughout, *transversal* means a tensor product of one-site physical operators with the party labels fixed. A *fixed-party* symmetry has this form; a *party-moving* symmetry additionally permutes tensor factors. LU and LC equivalence allow a party permutation when it is displayed explicitly. Logical groups called fixed-party below exclude such permutations.

Theorem 1.1 (Local-unitary rigidity). Let $q = p^e$, let $m \geq 2$, and let $|\psi\rangle, |\phi\rangle$ be stabilizer $\text{AME}(2m, q)$ states for the standard additive Weyl system. Suppose that, for a party permutation π , single-qudit unitaries $U_1, \dots, U_{2m}$, and a phase θ ,

$$(U_1 \otimes \cdots \otimes U_{2m})\pi|\psi\rangle = e^{i\theta}|\phi\rangle.$$

Then every U_i normalizes the finite-field Weyl system. In particular, the displayed equivalence is local Clifford.

The mechanism is visible in any $m+1$ -party reduced operator. Stabilizer labels supported on those parties form a group of order q^2 , and projection to the local Pauli-label group at every party in the marginal is bijective. The marginal is therefore diagonal on the complete local Weyl basis, with nonzero coefficients. A rank-one contraction recovers exactly one Weyl axis, so the tensor intrinsically determines every local Weyl frame. Product conjugation must permute those frames and is consequently Clifford. The argument uses neither CSS form, equal phases, classical linearity, nor $\mathbb{F}_q$ -linearity of the stabilizer labels. Thus the proof has four steps:

$$\begin{aligned} \text{AME support squeeze} &\implies \text{full-Weyl marginal} \\ &\implies \text{intrinsic local axes} \implies \text{local Clifford factors.} \end{aligned}$$

Corollary 1.2 (Transversal Clifford no-go). Let $m \geq 2$, and let $V_\psi, V_\phi : \mathbb{C}^q \rightarrow (\mathbb{C}^q)^{\otimes(2m-1)}$ be encoding isometries whose normalized Choi states are stabilizer $\text{AME}(2m, q)$ states $|\psi\rangle, |\phi\rangle$. They encode the associated $[[2m-1, 1, m]]_q$ quantum MDS codes. If

$$(U_1 \otimes \cdots \otimes U_{2m-1})V_\psi = V_\phi L$$

for a logical unitary L , then L and every U_i are Clifford. Thus every transversal conversion between two codes in the family is Clifford factor by factor. Taking $V_\psi = V_\phi$, no such code admits a transversal non-Clifford logical unitary.

The support bijections also retain how local Weyl frames on different parties fit together. On an $(m+1)$ -party support A , let $p_i : L_\psi(A) \rightarrow \mathbb{F}_q^2$ be the bijective projection at party i . The maps $M_A(i, j) = p_j p_i^{-1}$ push a Weyl label from i to j . Their collection over all A, i, j is the *minimum-support atlas*. Local trace-symplectic frame changes are compatible precisely when they intertwine every $M_A(i, j)$. Here a projective product-unitary group means the image of the product-unitary symmetries in the projective unitary group of the full tensor product. Schematically, overlapping supports turn operator pushing into a loop,

$$P_i \xrightarrow{M_A(i,j)} P_j \xrightarrow{M_B(j,k)} P_k \xrightarrow{M_C(k,i)} P_i,$$

whose composite is an atlas holonomy.

Theorem 1.3 (Minimum-support atlas classification). Let $m \geq 2$. After a fixed party relabelling, two stabilizer $\text{AME}(2m, q)$ states are LU equivalent exactly when their minimum-support operator-pushing atlases are equivalent by local symplectic changes of Weyl frame. For one state, let $\mathcal{G}_\psi$ be its group of compatible local symplectic gauges, let Γ_ψ be its fixed-party projective product-unitary symmetry group, and identify its projective Pauli stabilizer with the label group L_ψ . Then

$$1 \longrightarrow L_\psi \longrightarrow \Gamma_\psi \longrightarrow \mathcal{G}_\psi \longrightarrow 1$$

is exact. If q is prime, choosing base transports identifies $\mathcal{G}_\psi$ with the centralizer in $\mathrm{SL}_2(q)$ of the atlas holonomies.

Thus an a priori continuous LU-equivalence problem reduces to finite symplectic linear algebra on Weyl frames recovered from the state itself.

Rains recovered the three Pauli axes from a diagonal correlation tensor, and Van den Nest, Dehaene, and De Moor used this mechanism in a minimal-support LU-to-LC criterion for qubit stabilizer states [Rai99, Theorem 13] [VdNDDM05b, Theorem 1 and Corollary 1]. Corollary 3.6 gives a criterion of the same kind at every local dimension. Through each party, it asks for marginals of both states whose recovered Weyl labels generate the local label group. The states and coefficients are otherwise arbitrary: no stabilizer, CSS, or linearity hypothesis enters. For fully entangled qubit stabilizer states on at least three parties, the minimal-support lemma identifies this generation condition with the hypothesis of Van den Nest, Dehaene, and De Moor. The criterion therefore unifies their two proof branches rather than strengthening their theorem in local dimension two.

The distinction appears above local dimension two. The label group of a minimal support embeds in the local label group at each of its parties. In the qubit case it therefore has only the two nontrivial sizes just described; in larger dimension it may be intermediate. Such subgroups occur naturally in CSS states, where a minimum-weight X -support carrying no Z -word can contribute a line of labels. Proposition 3.10 verifies the stronger, full-Weyl hypothesis for stabilizer AME states over every prime power.

The axis-recovery linear algebra is classical [Har70, Kru77]. Chang and Jing introduced the general Kruskal-type tensor-uniqueness method into the qubit LU problem [CJ22]; their generic coefficient tensors yield LU invariants, not a Clifford conclusion. Only the additive label structure is used, so the criterion also applies to the integer-modulus Weyl system on $\mathbb{C}^d$ for arbitrary d . Section 3.1 gives the proof and its sharp two-party and nongenerating boundaries. Van den Nest, Dehaene, and De Moor also gave a finite invariant set characterizing LC equivalence of arbitrary binary stabilizer states [VdNDDM05a]. Our atlas theorem is not a first general LC-classification result: it uses the AME support profile to replace a global invariant list by an all-prime-power operator-pushing criterion and an explicit symmetry-group centralizer. Tan relates the LU orbits and local symmetries of even-party AME tensors to quantum-MDS codes and computes the full local symmetry group of the canonical four-qutrit AME state [Tan26, Theorems 3.5, 3.6, and 5.3]. The local factors in Tan’s displayed generators are qutrit Cliffords, so that calculation contains the $q = 3, m = 2$ automorphism subcase above. Our contributions are the uniform all-prime-power, all- m , arbitrary-additive intertwiner theorem and the exact minimum-support-atlas classification of its residual Clifford symmetries. The AME condition itself supplies both the full-Weyl marginal required by the axis argument and the minimum-support generation required by the atlas theorem.

Equivalently, the support bijection is a minimal-support Pauli form of perfect-tensor operator pushing: every one-party Weyl operator has a unique product-Weyl representative on any chosen m of the other parties. Perfect-tensor operator pushing and the associated $[[2m - 1, 1, m]]$ encoding interpretation were introduced in the holographic-code setting by Pastawski, Yoshida, Harlow, and Preskill [PYHP15, Eq. (2.3) and Section 5.7]. Overlapping minimum supports compare the recovered local Weyl-label frames. Theorem 1.3 makes their transition atlas a complete classification reduction rather than only an obstruction. For six parties, explicit holonomy conjugacy data recover the pencil coordinate z . The boundary $m \geq 2$ is sharp: a Bell pair has non-Clifford $U \otimes \bar{U}$ symmetries. Its pair marginal is pure rather than maximally mixed, which is exactly the failure that the second-moment computation of Section 5 detects.

Rigidity determines which product operations are possible; the maximum-distance-separable (MDS) Calderbank–Shor–Steane (CSS) model determines which Clifford operations

actually occur. If $C \leq \mathbb{F}_q^{2m}$ is a linear $[2m, m, m+1]_q$ MDS code, then

$$|\Psi_C\rangle = q^{-m/2} \sum_{c \in C} |c\rangle \quad (1.1)$$

is a stabilizer $\text{AME}(2m, q)$ state. Conversely, an equal-phase state of a linear $[2m, m]_q$ code is AME only if the code is MDS [RGRA18]. Here the MDS–CSS geometry is no longer needed for rigidity; it makes the exact fixed-party projective logical image computable.

Corollary 1.4 (Diagonal-isodual transversal logical group). Let q be an odd prime, let $m \geq 2$, and let $C \leq \mathbb{F}_q^{2m}$ be a linear $[2m, m, m+1]_q$ MDS code. The projective group of logical unitaries of the associated $[[2m-1, 1, m]]_q$ quantum MDS code that admit tensor-product physical implementations is

$$\begin{cases} \mathbb{F}_q^2 \rtimes \text{SL}_2(q), & \text{if } SC = C^\perp \text{ for a nonsingular diagonal } S, \\ \mathbb{F}_q^2 \rtimes T, & \text{otherwise,} \end{cases}$$

where $T = \{\text{diag}(a, a^{-1}) : a \in \mathbb{F}_q^\times\}$.

Put

$$\mathcal{D}(C, C^\perp) = \{s \in \mathbb{F}_q^{2m} : \text{diag}(s)C \subseteq C^\perp\}.$$

The branch condition is the nullity test $\dim \mathcal{D}(C, C^\perp) = 1$. Proposition 6.1 proves that this dimension is zero or one and that every nonzero vector in the space has full support and is unique up to scalar.

The nullity-one condition governs the two branches above. In odd characteristic the $\text{SL}_2(q)$ factor has coherent Weil lifts, while the affine projective group retains the usual Heisenberg central extension; Section 6 separates these lift questions. Generalized and extended generalized Reed–Solomon codes are diagonally isodual, but the condition is not confined to that family: self-dual non-GRS MDS constructions are known [ZL24].

For example, an extended $[8, 4, 5]_7$ Reed–Solomon code gives an $\text{AME}(8, 7)$ tensor and a $[[7, 1, 4]]_7$ quantum MDS code whose projective transversal logical group has order $7^2|\text{SL}_2(7)| = 16464$. No non-Clifford transversal gate enlarges this group.

For binary stabilizer codes, Wirthmüller determines the identity component of the local-unitary automorphism group and gives a protected-qubit criterion for finiteness of the induced automorphism group [Wir11, Theorem 7 and Corollary 11]. Our result instead determines every local factor inside the prime-power stabilizer-AME class.

Finiteness alone needs much less than either hypothesis. Section 5 proves that a pure state whose two-party marginals are all maximally mixed — any local dimension, no stabilizer structure, no maximal entanglement — has finite product-unitary symmetry group modulo phase, and that the defect of a product unitary grows quadratically away from that group, with an explicit ratio constant independent of the number of parties. This local estimate is stated in generator coordinates: the admissible ℓ^1 radius is governed by the state’s uniformity order, remaining bounded in the order-three family and growing linearly for an $\text{AME}(2m, q)$ state. It is not a defect ball. The stabilizer hypothesis then sharpens discreteness into uniform separation: because overlaps of stabilizer states are quantized, a product Clifford is either an exact symmetry or has defect at least $(2 - 2p^{-1/2})^{1/2}$, a constant depending on the characteristic and on nothing else. Rounding a near-symmetry to a Clifford and invoking that gap gives a defect-only threshold below which every approximate symmetry factors through an exact one. Reading one minimum-support marginal gives an exponentially small threshold, and aggregating all such supports improves only the low-dimensional range. The stronger route views each party as the logical leg of the exact $[[2m-1, 1, m]]_q$ code. A transversal physical implementation makes a nested logical Pauli commutator localize on one of three

Table 1: Scopes of the principal results. “Fixed after relabelling” means that a displayed party permutation is applied first and the remaining equivalence is fixed-party.

Result	Local dimension	State or code assumptions	Party motion
Discreteness and local stability	any integer	arbitrary 2-uniform pure state on $n \geq 4$ parties	fixed
LU rigidity	p^e	arbitrary stabilizer AME($2m, q$)	allowed explicitly
Atlas classification	p^e	arbitrary stabilizer AME($2m, q$)	fixed after relabelling
Exact fixed-party projective logical group	odd prime	equal-phase linear $[2m, m, m+1]$ MDS-CSS	fixed
Pencil LU/LC by z	odd prime	admitted non-GRS six-party pencil	arbitrary permutation

correctable regions; Fourier concentration then rounds the logical action to a Clifford. The exact cleanable-region obstruction is known [PY15, Lemma 5]; here it is made leakage-aware and quantitative before being combined with the stabilizer overlap gap. The resulting defect threshold has order $m^{-1/2}$ for every fixed local dimension, conditional on AME existence. This splits the rigidity phenomenon into an entanglement half, which forces discreteness, and an algebraic half, which names the finite group. It sharpens the limitation of the theorems above without removing it. Arbitrary phased minimal-support AME($6, d$) states still form infinitely many LU classes [RL25], and nonlinear orthogonal arrays and nonstabilizer AME states are still unclassified here; but every one of those states has a finite product-symmetry group modulo phase, so the stabilizer hypothesis is doing only the work of identifying which finite group appears.

The geometric applications in the remainder of the paper specialize to $m = 3$. Our main explicit application is the ordered pencil

$$H_t = ((0, 1, 1 - t), (0, 1, t - 1), (1, 1 - t, 0), (1, t - 1, 0), (1, 0, -t), (1, 0, t)). \quad (1.2)$$

On its admitted odd-field non-GRS locus, define

$$\begin{aligned} A(t) &= -4t(t - 1)^2, \\ B(t) &= (t^2 - t + 1)(t^2 - 3t + 1), \\ z(t) &= (B(t)/A(t))^2. \end{aligned} \quad (1.3)$$

Exact bracket identities show that z is the degree-eight projective quotient of the pencil. Theorem 1.1 upgrades its local-Clifford classification to the following complete local result.

Corollary 1.5 (Classification of the admitted pencil). Let t, u lie in the nondegenerate locus (7.1) over an odd prime field $\mathbb{F}_q$. Write $\sim_{\text{proj}}$ for projective equivalence of the ordered six-arcs up to a party permutation and $\sim_{\text{mon}}$ for monomial equivalence of their codes. Then

$$H_t \sim_{\text{proj}} H_u \iff \ker H_t \sim_{\text{mon}} \ker H_u \iff \Psi_t \sim_{\text{LC}} \Psi_u \iff \Psi_t \sim_{\text{LU}} \Psi_u \iff z(t) = z(u),$$

where party permutations are allowed.

The exact diagonal-isodual group dichotomy above holds for every m . For six parties it has a geometric coding-theoretic interpretation over odd prime fields. Regard any one tensor leg as the input of the associated $[[5, 1, 3]]_q$ encoder. The fixed-party product-Clifford symmetries induce the full logical $\text{SL}_2(q)$ precisely when the six-arc is conic/GRS; otherwise they induce only the split torus. Thus the nineteenth-century self-association condition for

six points controls which one-qudit logical Clifford gates are available without moving parties. Including logical Paulis, the exact fixed-party projective logical group is $\mathbb{F}_q^2 \rtimes \mathrm{SL}_2(q)$ on the GRS locus and $\mathbb{F}_q^2 \rtimes T$ off it.

Quantum Reed–Solomon and polynomial-code constructions provide the standard coding setting [GGB99, ABO08, Got99]. Closest to our fixed-party question are restrictions on diagonal gates between qubit codes [AJO14, Propositions 3 and 4], the endomorphism-algebra classification of multiblock diagonal Clifford gates [DB25, Theorems 5.5 and 6.1], and automorphism methods that recover SWAP-transversal logical Cliffords and their Pauli corrections [SKW⁺25]. Other locality models, including fold-transversal and 2-local layers, can realize larger groups [TCT26, Hol26]; fully transversal gates remain subject to the Eastin–Knill obstruction and, for codes with several logical qubits, cannot realize the full logical Clifford group [EK09, CG26].

Our coordinates may carry independent one-qudit factors, and Corollary 1.2 also treats conversions between different one-logical-qudit quantum-MDS encoders. The MDS–CSS specialization identifies the exact fixed-party projective image for every half-dimensional MDS code over an odd prime field. Its dividing condition, diagonal code–dual equivalence, also occurs beyond GRS codes in algebraic-geometry MDS families [ZZ26].

Classically, we use Coble’s association theory [Cob22] through the modern treatment of Dolgachev and Ortland [DO88, Chapter III]; for finite-projective arc background see Segre, Hirschfeld, and the modern survey of Ball and Lavrauw [Seg55, Hir98, BL19]. Over characteristic zero, the GIT quotient of six ordered points in $\mathbb{P}^2$ is a double cover of $\mathbb{P}^4$, with Gale association exchanging the sheets and the self-associated locus forming the Igusa-quartic branch divisor [HMSV08, Section 2.3]. That divisor is dual to the Segre cubic parametrizing six points on $\mathbb{P}^1$. Our six-arc phase theorem is the finite-field, fixed-label slice of this picture: the branch condition becomes diagonal isoduality, and the Veronese determinant detects the conic/GRS locus. This interpretation is not used in the proof.

Sections 2–4 contain the main line: the arc-code–AME conventions, LU-to-LC rigidity, and its structural completion by the minimum-support atlas. Section 5 develops discreteness and quantitative stability, while Section 6 determines the exact MDS–CSS logical images and their lifts. The next three sections develop the six-party applications and contrasts. Section 7 proves the z -classification; Section 8 proves the logical-Clifford phase theorem; and Section 9 compares operator-valued rigidity with the generic blindness of fixed-copy scalar invariants. Section 10 records the trust boundary, and Section 11 closes with the exact extension-field boundary. The appendices contain the detailed H_3 and $q = 13$ scalar certificates, optional transport calculations, and finite party-permutation examples; none is used in the uniform rigidity or atlas theorems.

2 Stabilizer AME states and MDS–CSS models

Let $q = p^e$ and fix the additive character $\chi(a) = \exp(2\pi i \operatorname{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(a)/p)$. On the standard basis of $\mathbb{C}^q$, indexed by $\mathbb{F}_q$, put

$$X(a)|x\rangle = |x + a\rangle, \quad Z(b)|x\rangle = \chi(bx)|x\rangle.$$

For $v = (a, b) \in \mathbb{F}_q^2$, write $W_v = X(a)Z(b)$, with any fixed consistent phase convention when only its projective axis is used. The q^2 single-party operators W_v form an orthogonal basis of the single-party Hilbert–Schmidt space. A unitary is Clifford when conjugation permutes their projective axes CW_v .

After choosing dual $\mathbb{F}_p$ -bases of $\mathbb{F}_q$, this Weyl system is the Pauli system of e p -level factors. Its projective Clifford normalizer has the standard exact sequence

$$1 \longrightarrow \mathbb{F}_p^{2e} \longrightarrow \mathrm{PCliff}(q) \longrightarrow \mathrm{Sp}_{2e}(\mathbb{F}_p) \longrightarrow 1, \quad (2.0)$$

including $p = 2$: the kernel is the projective Pauli group, and every trace-symplectic $\mathbb{F}_p$ -linear label map has a Clifford lift [DDM03, HDDM05].

For n parties, a pure stabilizer state is fixed by an abelian Pauli group $\mathcal{S}$ of order q^n . Its projective label group $L \leq (\mathbb{F}_q^2)^n$, viewed additively over $\mathbb{F}_p$, is maximal isotropic for the trace-symplectic form. This is the standard arbitrary-additive prime-power stabilizer correspondence [KKKS06, Theorems 13 and 15]. For each $\ell \in L$, write the stabilizer lift as $\widetilde{W}_\ell = \alpha_\ell W_\ell$, where $\alpha_\ell \neq 0$. Then

$$|\psi\rangle\langle\psi| = q^{-n} \sum_{\ell \in L} \widetilde{W}_\ell. \quad (2.1)$$

If A is a party set, put

$$L(A) = \{\ell \in L : \text{supp}(\ell) \subseteq A\}.$$

Weyl orthogonality and partial trace give

$$\rho_A^\psi = q^{-|A|} \sum_{\ell \in L(A)} \widetilde{W}_\ell. \quad (2.2)$$

In particular, ρ_A^ψ is maximally mixed exactly when $L(A) = \{0\}$. Thus a stabilizer state on $2m$ parties is AME exactly when it has no nonidentity stabilizer label supported on at most m parties.

The rigidity and atlas theorems use only (2.0)–(2.2). For the geometric classification and explicit six-party applications, we now specialize to the MDS–CSS model.

A six-arc is an ordered sextuple of points of $\mathbb{P}^2(\mathbb{F}_q)$ with no three collinear. Choose nonzero column representatives and collect them in a 3-by-6 matrix H . The arc condition says that every three-column minor of H is nonzero. Consequently

$$C = \ker H \leq \mathbb{F}_q^6$$

is an $[6, 3, 4]_q$ MDS code, and its dual $C^\perp = \text{rowspan } H$ is MDS with the same parameters. Conversely, a parity-check matrix of such a code supplies a six-arc. Projectivities, independent rescaling of columns, and permutation of columns correspond exactly to monomial equivalences of the kernels. These objects exist exactly when $q \geq 4$. Indeed, from one point of a six-arc the five joining lines are distinct, so $q + 1 \geq 5$. For $q = 4$, a conic together with its nucleus is a six-arc; for $q \geq 5$, any six points of a nonsingular conic form a six-arc.

For the resulting CSS stabilizer we use the exact lift

$$W_{(c,h)} = X(c)Z(h) = \bigotimes_{i=1}^6 X(c_i)Z(h_i), \quad (c, h) \in C \oplus C^\perp.$$

Since $h \cdot c' = 0$ for $h \in C^\perp$ and $c' \in C$, these lifts multiply without a residual phase and fix $|\Psi_C\rangle$.

Proposition 2.1 (Arc–AME dictionary). For a six-arc with kernel C , the state $|\Psi_C\rangle$ in (1.1) is a minimal-support stabilizer AME(6, q) state. Its projective Pauli-label Lagrangian is

$$L_C = (C \oplus 0) + (0 \oplus C^\perp) \leq \mathbb{F}_q^{12}.$$

Monomial code equivalence, together with a party permutation, induces local-Clifford equivalence of the corresponding states.

Proof. The operators $X(c)$, $c \in C$, fix the equal superposition. The operators $Z(h)$, $h \in C^\perp$, fix it because $\chi(h \cdot c) = 1$. The two subgroups commute, and their label space has dimension six and is Lagrangian.

For any three-coordinate set I , puncturing C to I is bijective: injectivity follows from distance four and both spaces have dimension three. In the partial trace of $|\Psi_C\rangle\langle\Psi_C|$, equality on the complementary three coordinates therefore forces equality of the codewords, while the projection to I runs once through $\mathbb{F}_q^3$. Hence every three-party reduction is $q^{-3}I$. Smaller reductions are maximally mixed by tracing again, so the state is $\text{AME}(6, q)$. Its support has the minimum possible size q^3 .

A column multiplier $x \mapsto ax$ is a single-party Clifford operation: it sends $X(b)$ to $X(ab)$ and $Z(b)$ to $Z(a^{-1}b)$. Projective row operations do not change the kernel, and column permutations permute parties. This proves the last assertion. $\square$

For a party set S , let

$$L_C(S) = \{\ell \in L_C : \text{supp}(\ell) \subseteq S\}.$$

The general stabilizer partial-trace formula specializes to

$$\rho_S^C = q^{-|S|} \sum_{\ell \in L_C(S)} W_\ell. \quad (2.3)$$

For later block calculations, write

$$C_X = C \oplus 0, \quad C_Z^\perp = 0 \oplus C^\perp, \quad L_C = C_X \oplus C_Z^\perp.$$

This formula is the bridge used in both the rigidity proof and the marginal invariants below. A Pauli-labelled quantity is not by itself an LU invariant; the invariant statements will always be expressed either as covariance of the reduced operators or as basis-free tensor contractions.

The GRS boundary has a simple geometric meaning. A dimension-three length-six generalized Reed–Solomon code is represented, up to column multipliers, by six points on a nonsingular conic in $\mathbb{P}^2$. We use “GRS state” for the state of any such code. The non-GRS pencil in Section 7 consists of arc matrices H_t away from that conic locus and from the other factors needed to keep the columns distinct and the displayed coordinates defined.

3 Full-Weyl marginals and LU rigidity

We first isolate the linear-algebra statement that recovers a diagonal tensor’s coordinate axes. It is the full-Weyl counterpart of the three-Pauli argument in [Rai99, VdNDDM05b]. The statement itself is classical: it is Jennrich’s uniqueness theorem for trilinear decompositions, published in [Har70, §V] and there credited to Jennrich, whose conclusion that the factor matrices agree up to one permutation and compensating diagonal scalings is the monomial property below. It is the full-column-rank case of Kruskal’s condition [Kru77], extended to arbitrarily many factors in [SB00]; see [KB09, §3.2] for the attribution and for a survey of the uniqueness theory. We give a short self-contained proof because the contraction argument is the one used in Proposition 3.2, and claim no novelty for either the statement or the proof.

Lemma 3.1 (Diagonal-tensor axes). Let $r \geq 3$, let $E_1, \dots, E_r$ be N -dimensional complex inner product spaces, and choose orthonormal bases $\{e_{ij} : 1 \leq j \leq N\}$. If

$$T = \sum_{j=1}^N \lambda_j e_{1j} \otimes \cdots \otimes e_{rj}, \quad \lambda_j \neq 0, \quad (3.1)$$

then the coordinate axes in every E_i are intrinsic to T . Consequently, a product of invertible linear maps carrying one tensor of the form (3.1) to another is monomial in the displayed bases.

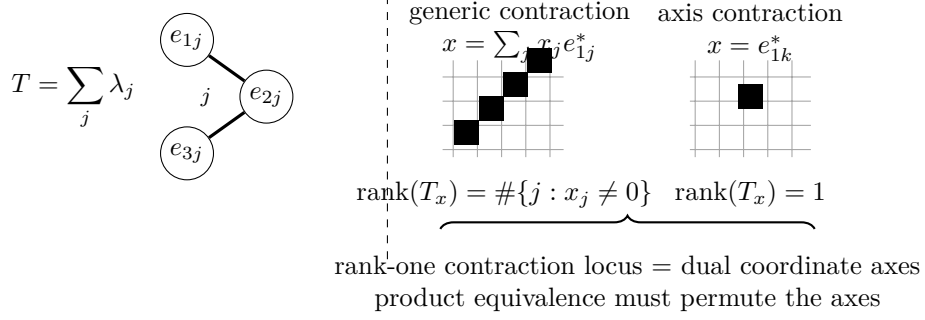


Figure 1: A diagonal tensor with all coefficients nonzero exposes its own axes. Contracting one leg against x and flattening the remainder gives rank equal to the number of nonzero coordinates of x , so the rank-one contraction locus is exactly the set of dual coordinate axes. For an $(m + 1)$ -party AME marginal in local Hilbert–Schmidt spaces, the recovered axes are the local Weyl axes, and a product conjugation must permute them.

Proof. For nonzero $x = \sum_j x_j e_{1j}^*$, contract the first factor against x . The result is

$$T_x = \sum_j \lambda_j x_j e_{2j} \otimes \cdots \otimes e_{rj}.$$

Flattening from E_2 to $E_3 \otimes \cdots \otimes E_r$ gives a matrix of rank $\#\{j : x_j \neq 0\}$. Thus T_x has rank one exactly when the ray of x is a dual coordinate axis. Product equivalence preserves this projective rank-one contraction locus. The inner products identify the recovered dual axes with the coordinate axes in the first factor. Apply the same argument in every factor. $\square$

Call an operator on $r \geq 3$ qudits *full-Weyl diagonal* if there is a q^2 -element set P , a distinguished element $0 \in P$, and bijections $f_i : P \rightarrow \mathbb{F}_q^2$ with $f_i(0) = 0$, for which

$$R = \sum_{v \in P} \lambda_v \bigotimes_{i=1}^r W_{f_i(v)}, \quad \lambda_v \neq 0. \quad (3.2)$$

The coefficients need not be equal. What matters is that the same q^2 -element index set runs bijectively through the complete local Weyl basis at every party. Neither P nor the bijections need be $\mathbb{F}_q$ -linear.

Proposition 3.2 (Full-Weyl marginal criterion). Let R and R' be full-Weyl diagonal operators on the same $r \geq 3$ parties, possibly with different indexing sets, bijections, and coefficients. If

$$R' = (U_1 \otimes \cdots \otimes U_r) R (U_1 \otimes \cdots \otimes U_r)^\dagger,$$

then every U_i is Clifford.

Proof. View R and R' as tensors in the local Hilbert–Schmidt spaces. After replacing W_v by the orthonormal basis $q^{-1/2} W_v$ and absorbing the common normalization into the coefficients, both have the form in Lemma 3.1, with $N = q^2$. Local conjugation acts invertibly on each factor, so it must permute the Weyl axes. It fixes the identity operator and hence the identity axis. It therefore permutes the nonidentity projective Weyl operators, which is exactly the single-qudit Clifford condition. $\square$

Corollary 3.3 (Full-Weyl marginal-cover rigidity). Let $|\psi\rangle, |\phi\rangle$ be n -qudit states and suppose a product unitary carries $|\psi\rangle$ to $|\phi\rangle$ up to phase, any party permutation having first been absorbed by relabelling $|\phi\rangle$, so that the two states are indexed by the same parties. If every

party belongs to a set S of at least three parties for which both reduced operators ρ_S^ψ and ρ_S^ϕ are full-Weyl diagonal, then every local factor is Clifford.

Proof. Partial trace gives a product-unitary equivalence of the two reduced operators on each chosen S . Apply Proposition 3.2; the chosen sets cover all parties. $\square$

3.1 Partial marginals and the recognition group

The full-Weyl criterion recovers the entire local Weyl basis from one marginal. More generally, several marginals may each recover only part of that basis, while their labels together generate it. The following criterion isolates this mechanism. It uses only a finite abelian label group and a Weyl basis whose multiplication is additive up to phase.

Call an operator on r parties *partial-Weyl diagonal* if there are a finite index set P with a distinguished element 0, injections $f_i : P \rightarrow \mathbb{F}_q^2$ with $f_i(0) = 0$, and nonzero coefficients, for which

$$R = \sum_{v \in P} \lambda_v \bigotimes_{i=1}^r W_{f_i(v)}, \quad \lambda_v \neq 0.$$

Write $N = |P|$. The full-Weyl condition (3.2) is the case $N = q^2$, where the injections are forced to be bijections.

Proposition 3.4 (Partial-Weyl marginal criterion). Let R and R' be partial-Weyl diagonal operators on the same r parties with index sets of the same size $N \geq 2$, possibly with different index sets, injections, and coefficients, and suppose

$$R' = (U_1 \otimes \cdots \otimes U_r) R (U_1 \otimes \cdots \otimes U_r)^\dagger.$$

Assume $r \geq 3$, or $N = 2$. Then for each i , conjugation by U_i carries every $W_{f_i(v)}$ to a phase times a Weyl operator, and induces a bijection $f_i(P) \rightarrow f'_i(P')$ fixing 0.

Proof. Suppose $r \geq 3$. Let E_i be the span of $\{W_{f_i(v)} : v \in P\}$ inside the single-party Hilbert–Schmidt space and E'_i the span of $\{W_{f'_i(v')} : v' \in P'\}$. Both have dimension N , because the Weyl operators are linearly independent and the injections are injective, and the rescaled operators $q^{-1/2}W_v$ are orthonormal. All N coefficients of R are nonzero, so by the rank computation in the proof of Lemma 3.1 the smallest subspace of factor i supporting R is E_i , and likewise for R' and E'_i ; hence the product of the conjugations carries E_i onto E'_i . Now apply Lemma 3.1 to the two tensors in $E_1 \otimes \cdots \otimes E_r$ and $E'_1 \otimes \cdots \otimes E'_r$. Its proof uses only $r \geq 3$, nonzero coefficients, and orthonormal bases, never $N = q^2$. The maps are therefore monomial in the displayed bases, and each scalar is a phase because conjugation is unitary for the Hilbert–Schmidt inner product. The identity axis is fixed, so the bijection fixes 0.

Suppose instead $N = 2$, with $R = \lambda_0 I + \lambda_1 \bigotimes_i W_{f_i(1)}$ and $R' = \lambda'_0 I + \lambda'_1 \bigotimes_i W_{f'_i(1)}$. Conjugation fixes the identity, so comparing traces gives $\lambda_0 = \lambda'_0$; subtracting leaves an equality of nonzero product operators, whose tensor factorization gives $U_i W_{f_i(1)} U_i^\dagger = c_i W_{f'_i(1)}$ with $\prod_i c_i = \lambda'_1 / \lambda_1$, and each c_i is a phase by unitarity. $\square$

The arity condition cannot be removed in this form. For two parties and three or more terms, equal coefficient moduli leave a unitary freedom among the corresponding Schmidt axes. With two terms, conjugation fixes the identity term and hence singles out the other axis.

Fix a product unitary carrying $|\psi\rangle$ to $|\phi\rangle$ up to phase, after absorbing any party permutation by relabelling $|\phi\rangle$. For each party i put

$$\Lambda_i = \{a \in \mathbb{F}_q^2 : U_i W_a U_i^\dagger \in \mathbb{C}^\times W_b \text{ for some } b\},$$

the *recognition group* at i .

Lemma 3.5 (Recognition group). Λ_i is a subgroup of $\mathbb{F}_q^2$, the induced map $a \mapsto b$ is an injective additive homomorphism, and U_i is Clifford if and only if $\Lambda_i = \mathbb{F}_q^2$.

Proof. If $U_i W_a U_i^\dagger = \alpha W_{a'}$ and $U_i W_b U_i^\dagger = \beta W_{b'}$, then $W_a W_b$ is a phase times W_{a+b} , so conjugating that product shows $U_i W_{a+b} U_i^\dagger$ is a scalar times $W_{a'+b'}$. Hence Λ_i is closed under addition and the induced map is additive; it is injective because distinct labels have distinct projective axes. If $\Lambda_i = \mathbb{F}_q^2$ the induced map is an injective endomorphism of a finite group, hence bijective, so conjugation permutes all projective Weyl axes, which is the Clifford condition. The converse is immediate. $\square$

When $q = 2$, a recognition group containing a nonzero label is either a line or all of $\mathbb{F}_2^2$. The criterion below assembles labels recovered from different marginals; in larger local dimension it also accommodates intermediate subgroups that have no qubit analogue.

Corollary 3.6 (Generation criterion). Suppose that for each party i there is a family of party sets containing i , each with at least three parties or with a two-element index set, such that ρ_S^ψ and ρ_S^ϕ are partial-Weyl diagonal with index sets of equal size. If the labels recovered at i over that family generate $\mathbb{F}_q^2$, then U_i is Clifford.

Proof. Partial trace gives a product-unitary equivalence of the two reduced operators on each set. Proposition 3.4 places each recovered label in Λ_i , and Lemma 3.5 closes Λ_i under addition and converts $\Lambda_i = \mathbb{F}_q^2$ into the Clifford property. $\square$

Corollary 3.3 is the case of one set per party with $N = q^2$, where generation is automatic. For stabilizer states the supply of usable sets is exactly the minimal supports.

Lemma 3.7 (Minimal supports carry partial-Weyl marginals). Let $|\psi\rangle$ be a stabilizer state and ω a minimal support of its stabilizer, and let $L(\omega)$ be the group of labels supported inside ω . Then for every $i \in \omega$ the coordinate projection $\pi_i : L(\omega) \rightarrow \mathbb{F}_q^2$ is injective. Hence $L(\omega)$ is isomorphic to a subgroup of $\mathbb{F}_q^2$, every nonzero element of $L(\omega)$ has support exactly ω , and ρ_ω is partial-Weyl diagonal with index set $L(\omega)$ and injections π_i . Conversely, if the retained label group on a party set has at least two elements and injects at every party, that set is a minimal support.

Proof. An element of the kernel of π_i has support contained in $\omega \setminus \{i\}$, a proper subset of ω , so it is zero by minimality. The partial trace of $|\psi\rangle\langle\psi|$ over the complement of ω retains exactly the stabilizer elements supported inside ω , each with a coefficient of modulus one. For the converse, injectivity at every party of the set says that no nonzero element of its label group vanishes anywhere on it, so no nonzero element has smaller support. $\square$

At $q = 2$ the possible orders of $L(\omega)$ are 2 and 4, which is the dichotomy of [VdNDDM05b, Lemma 1]; the intermediate orders available for $q > 2$ are invisible there. They occur naturally in CSS states: a minimum-weight X -support carrying no Z -word contributes a line of q labels.

For two LU-equivalent stabilizer states, marginal purity preserves both the minimal supports and the order of each retained label group, since

$$\text{Tr}(\rho_\omega^2) = \frac{|L(\omega)|}{q^{|\omega|}}.$$

Thus every source chart above has a target chart of the same size. In the qubit case, the hypothesis of [VdNDDM05b, Theorem 1] says exactly that the labels from all minimal supports through a party generate $\mathbb{F}_2^2$. Label groups of order two use the two-term case of Proposition 3.4; label groups of order four use its $r \geq 3$ case. Their standing assumptions of at least three parties and full entanglement exclude a two-party support in the latter case. Corollary 3.6 therefore recovers their full theorem, not only its single-support corollary.

Corollary 3.8 (CSS states over a prime field). Let q be prime and let $|\psi\rangle$ be a CSS coset state such that every coordinate lies in two minimal supports of the stabilizer, each with at least three coordinates, one whose label group contains a nonzero X -type label and one whose label group contains a nonzero Z -type label. Then every product-unitary equivalence to another such state has Clifford factors.

Proof. The preceding purity argument supplies matching partial-Weyl marginals for the target state. The two supports contribute independent labels $(x, 0)$ and $(0, z)$ to Λ_i , which generate $\mathbb{F}_q^2$ over a prime field. Apply Corollary 3.6. $\square$

A minimum-weight codeword of the X code need not give a minimal support of the stabilizer, since a Z -word may sit strictly inside it; the hypothesis above is about minimal supports of the full stabilizer. What is not automatic for an arbitrary additive stabilizer is either coverage by minimal supports or the availability of both label types at every party; the corollary keeps both requirements explicit.

Remark 3.9 (Arbitrary local dimension). Nothing in the partial-Weyl condition, Proposition 3.4, Lemma 3.5, Corollary 3.6, or Lemma 3.7 uses the field structure of $\mathbb{F}_q$. Each uses only that the labels form a finite abelian group whose Weyl operators are an orthogonal Hilbert–Schmidt basis multiplying additively up to phase. Replacing $\mathbb{F}_q^2$ by $\mathbb{Z}_d^2$, with X the shift and Z the clock operator on $\mathbb{C}^d$, all five statements hold verbatim at every local dimension d , prime power or not. At composite d , a unitary whose conjugation permutes the projective Weyl axes still normalizes the Pauli group itself and is Clifford in the standard sense. Write $UW_aU^\dagger = cW_{a'}$. For odd d , every Weyl operator satisfies $W_a^d = I$; taking d -th powers gives $c^d = 1$. For even d , every Weyl operator satisfies $W_a^{2d} = I$, and the standard Pauli convention includes the $2d$ -th roots of unity; taking $2d$ -th powers gives $c^{2d} = 1$. Thus c is already a Pauli phase and $UW_aU^\dagger$ lies in the Pauli group. When d is a proper prime power, the $\mathbb{Z}_d$ Weyl system and the $\mathbb{F}_d$ Weyl system on $\mathbb{C}^d$ are different, and a criterion instantiated in one says nothing in the other.

The generation hypothesis is sufficient, not necessary. Wong and Jiang exhibit at $d = 9$ a three-party stabilizer state that is LU but not LC equivalent to the GHZ state [WJ25]. Its three two-party minimal supports each carry nine labels whose projections lie in $3\mathbb{Z}_9 \times 3\mathbb{Z}_9$. They generate a subgroup of index nine rather than $\mathbb{Z}_9^2$, so Corollary 3.6 does not apply.

Chang and Jing introduced the general Kruskal-type uniqueness method for coefficient tensors in a discrete operator basis into the qubit LU problem [CJ22]. Their generic Pauli tensors are not diagonal in that basis, and uniqueness yields invariants separating LU classes rather than a Clifford conclusion. For qubit stabilizer states that conclusion is [VdNDDM05b, Theorem 1].

Proposition 3.10 (Stabilizer-AME support theorem). Let $|\psi\rangle$ be a stabilizer AME($2m, q$) state, and let A be any $m + 1$ parties. Then

$$|L(A)| = q^2.$$

Every nonidentity label in $L(A)$ has support exactly A , and for each $i \in A$ the coordinate projection

$$p_i : L(A) \longrightarrow \mathbb{F}_q^2$$

is bijective.

Proof. The projective label group L has order q^{2m} . Restrict its labels to the $m - 1$ parties outside A . The target has order $q^{2(m-1)}$, and the kernel is $L(A)$; hence

$$|L(A)| \geq q^{2m}/q^{2(m-1)} = q^2. \quad (3.3)$$

Fix $i \in A$. A label in the kernel of p_i is supported on the m -set $A \setminus \{i\}$. The AME condition and (2.2) exclude every such nonidentity stabilizer, so p_i is injective. Its codomain has order q^2 , giving the reverse inequality in (3.3) and hence bijectivity. The same no-small-support argument shows that every nonidentity element of $L(A)$ has full support A . $\square$

Proposition 3.11 (A stabilizer-AME marginal fixes the local Weyl axes). Let $|\psi\rangle, |\phi\rangle$ be stabilizer AME($2m, q$) states. If a product unitary on an $(m+1)$ -set A carries ρ_A^ψ to ρ_A^ϕ , then every local factor on A is Clifford.

Proof. By (2.2),

$$\rho_A^\psi = q^{-|A|} \sum_{\ell \in L_\psi(A)} \widetilde{W}_\ell.$$

Index this sum by $P = L_\psi(A)$. Proposition 3.10 says that every coordinate projection $P \rightarrow \mathbb{F}_q^2$ is bijective. Relative to the fixed Weyl convention, each stabilizer lift contributes a nonzero coefficient. Thus ρ_A^ψ is full-Weyl diagonal in the sense of (3.2), and the same holds for ρ_A^ϕ . Apply Proposition 3.2. $\square$

Proof of Theorem 1.1. Absorb the party permutation by relabelling $|\phi\rangle$. Partial trace gives a product-unitary equivalence $\rho_A^\psi \sim \rho_A^\phi$ for every $(m+1)$ -set A . Given a party i , choose such a set containing it and apply Proposition 3.11. Thus every U_i is Clifford. $\square$

4 Minimum-support atlases and encoder consequences

The same support calculation recovers standard quantum-MDS structure. If $|A| \leq m$, the AME condition gives $L(A) = \{0\}$. If $|A| > m$, then $\rho_{A^c}^\psi$ is maximally mixed, so purity of the global state gives $\text{Tr}((\rho_A^\psi)^2) = q^{-(2m-|A|)}$. On the other hand, (2.2) and Weyl orthogonality give $\text{Tr}((\rho_A^\psi)^2) = q^{-|A|}|L(A)|$. Hence, for every party set A ,

$$|L(A)| = q^{2 \max(|A|-m, 0)}. \quad (4.1)$$

Möbius inversion is the stabilizer-state specialization of the quantum-MDS Shor–Laflamme distribution of Huber and Grassl [HG20, Theorems 8 and 9]; in particular the number of weight- $m+1$ projective stabilizers is $\binom{2m}{m+1}(q^2-1)$. We use one further consequence. If $|A| \geq m+2$ and $i \neq j$ lie in A , then the dimensions in (4.1), together with

$$L(A \setminus \{i\}) \cap L(A \setminus \{j\}) = L(A \setminus \{i, j\}),$$

give

$$L(A) = L(A \setminus \{i\}) + L(A \setminus \{j\}). \quad (4.2)$$

Descending induction shows that the $(m+1)$ -supported subgroups span the full label group. This is the additive quantum-MDS form of minimum-weight generation, not a separate weight-enumerator claim.

We now prove the atlas classification stated in the introduction. Regard each local label group

$$P_i = \mathbb{F}_q^2$$

as a $2e$ -dimensional trace-symplectic space over $\mathbb{F}_p$. For every $(m+1)$ -set A and $i, j \in A$, the support theorem defines the operator-pushing transition

$$M_A(i, j) = p_j p_i^{-1} : P_i \longrightarrow P_j.$$

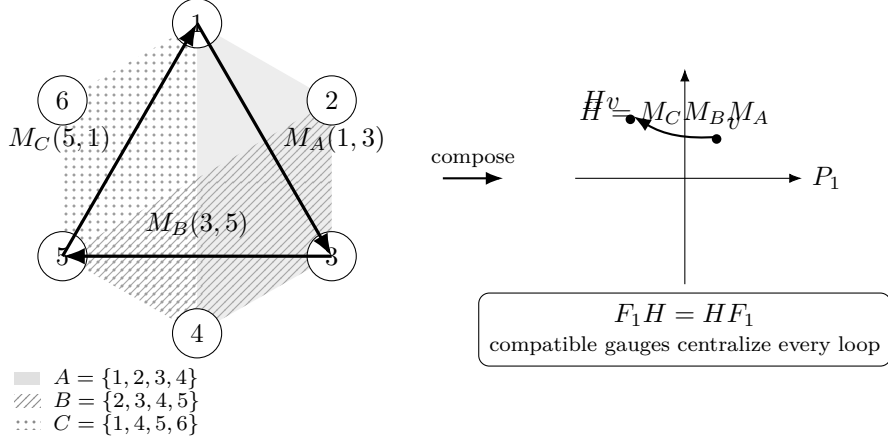


Figure 2: Operator pushing on minimum supports, shown for six parties. Each $(m+1)$ -party support identifies the local Weyl-label planes of its parties through the bijective projections of the support theorem; overlapping supports compose to a loop whose holonomy is a linear map of the base plane. An atlas gauge is compatible precisely when it intertwines every transition; for one state over a prime field, its base block centralizes every loop holonomy.

For two states ψ, ϕ on the same labelled parties, a *symplectic atlas equivalence* is a tuple $F_i \in \text{Sp}(P_i)$ satisfying

$$F_j M_A^\psi(i, j) = M_A^\phi(i, j) F_i \quad \text{for every } A \text{ and } i, j \in A. \quad (4.3)$$

Here $\text{Sp}(P_i) = \text{Sp}_{2e}(\mathbb{F}_p)$ is the action induced by the standard additive local Clifford group.

For one state, define

$$\begin{aligned} \Gamma_\psi &= \{[U] \in \text{PU}(q^{2m}) : U = U_1 \otimes \cdots \otimes U_{2m}, U|\psi\rangle \in \mathbb{C}|\psi\rangle\}, \\ \mathcal{G}_\psi &= \{(F_i) \in \prod_i \text{Sp}(P_i) : F_j M_A(i, j) = M_A(i, j) F_i \text{ for all } A, i, j \in A\}. \end{aligned}$$

Theorem 1.1 and the Clifford sequence (2.0) give a well-defined homomorphism

$$\lambda_\psi : \Gamma_\psi \longrightarrow \mathcal{G}_\psi, \quad [U_1 \otimes \cdots \otimes U_{2m}] \longmapsto (\overline{U}_i)_i,$$

where $\overline{U}_i$ is the induced symplectic label map. A product decomposition is unique up to one-site scalars, which do not change $\overline{U}_i$. The injection $L_\psi \hookrightarrow \Gamma_\psi$ sends a label ℓ to the projective Pauli $[W_\ell]$.

Lemma 4.1 (Pauli phase correction). Let ψ and ϕ be stabilizer states with Lagrangian label groups L_ψ and L_ϕ . If a product Clifford U induces a symplectic label map carrying L_ψ onto L_ϕ , then there is a product Pauli W_v such that $W_v U|\psi\rangle \in \mathbb{C}|\phi\rangle$.

Proof. The state $U|\psi\rangle$ is stabilized on the same Pauli axes as $|\phi\rangle$, possibly with a different stabilizer character. The trace-symplectic pairing identifies the ambient label space modulo $L_\phi = L_\phi^\perp$ with the character dual of L_ϕ . Choose v whose pairing character is the inverse of the discrepancy. Conjugation by the product Pauli W_v cancels it. The corrected state and $|\phi\rangle$ therefore lie in the same one-dimensional joint stabilizer eigenspace. $\square$

Proof of Theorem 1.3. First fix the party labels. A block symplectic map $F = \bigoplus_i F_i$ carrying L_ψ to L_ϕ carries every supported subgroup $L_\psi(A)$ to $L_\phi(A)$. Writing these subgroups as graphs through any projection p_i gives exactly (4.3).

Conversely, suppose (4.3) holds. Fix A and $i \in A$. Every element of $L_\psi(A)$ is uniquely determined by its i -coordinate, and the equations with $j \in A$ show that its image under F is the unique element of $L_\phi(A)$ with i -coordinate $F_i p_i(\ell)$. Thus $FL_\psi(A) = L_\phi(A)$. Since the $(m+1)$ -supported subgroups generate the full label groups, $FL_\psi = L_\phi$.

Sequence (2.0) supplies local Clifford lifts of the F_i , and Lemma 4.1 supplies the product-Pauli correction. Hence an atlas equivalence lifts to an LC equivalence. The reverse implication follows from the induced label action, and Theorem 1.1 replaces LC by LU. A party permutation is absorbed by relabelling first.

For automorphisms, λ_ψ is surjective by the same lifting and phase-correction argument. Its kernel consists of projective product Paulis by (2.0). Such a Pauli fixes the state ray exactly when its label lies in $L_\psi^\perp = L_\psi$. Thus the displayed injection of L_ψ identifies it with $\ker \lambda_\psi$, proving exactness.

Finally suppose $q = p$. Form the directed multigraph with vertices the parties and an edge (A, i, j) carrying $M_A(i, j)$. Every pair of parties belongs to an $(m+1)$ -set, so the graph is connected. Choose a base b , a spanning tree, and tree transports $T_i : P_b \rightarrow P_i$, with $T_b = I$. For every edge define its based holonomy

$$H_A(i, j) = T_j^{-1} M_A(i, j) T_i \in \text{GL}_2(q).$$

Compatibility propagates a base block F_b uniquely as $F_i = T_i F_b T_i^{-1}$; the edge (A, i, j) is then compatible exactly when F_b commutes with $H_A(i, j)$. Since $\text{Sp}_2(\mathbb{F}_q) = \text{SL}_2(q)$ is normal in $\text{GL}_2(q)$, every propagated block remains symplectic. Evaluation at the base therefore identifies $\mathcal{G}_\psi$ with the holonomy centralizer inside $\text{SL}_2(q)$. Different spanning trees generate the same based holonomy subgroup; changing the base conjugates that subgroup, and hence its centralizer, by a chosen transport. $\square$

4.1 Encoder consequences

Let $C \leq \mathbb{F}_q^{2m}$ be $[2m, m, m+1]_q$ MDS, and fix a party set S of size $m+1$. Shortening C on the $m-1$ coordinates outside S gives an $[m+1, 1, m+1]_q$ MDS code. The dual $C^\perp$ has the same parameters, so its corresponding shortening does as well. Choose nonzero shortened words $c^S \in C$ and $h^S \in C^\perp$. Each has support exactly S , and

$$L_C(S) = \{(ac^S, bh^S) : a, b \in \mathbb{F}_q\}. \quad (4.4)$$

At a party $i \in S$, projection of this plane to the local Pauli labels is

$$(a, b) \mapsto (ac_i^S, bh_i^S),$$

which is invertible because both coordinates are nonzero. Thus all q^2 labels in $L_C(S)$ run bijectively through the full local Weyl basis at every party of S ; in particular, the $q^2 - 1$ nonidentity labels run bijectively through the nonidentity Weyl labels.

Proof of Corollary 1.2. The input marginal of each AME Choi state is $q^{-1}I$, so one-leg reshaping gives an isometry. The projection of its stabilizer label group onto the input Weyl-label space is surjective: otherwise a nonzero local label orthogonal to the image would lie in the maximal isotropic space $L^\perp = L$, contradicting the absence of a one-party stabilizer. The kernel therefore has q^{2m-2} elements and stabilizes the image, making it a one-logical-qudit stabilizer code. AME corrects erasure of any $m-1$ outputs, so its distance is at least m ; the quantum Singleton bound gives equality. This is the standard perfect-tensor/quantum-MDS correspondence used in [Tan26, Theorems 3.5 and 3.6].

Let $|\Phi\rangle = q^{-1/2} \sum_x |x, x\rangle$, and use the Choi convention $|\psi\rangle = (I \otimes V_\psi)|\Phi\rangle$ and $|\phi\rangle = (I \otimes V_\phi)|\Phi\rangle$. Writing $U_{\text{phys}} = \bigotimes_i U_i$, the relation $U_{\text{phys}} V_\psi = V_\phi L$ gives

$$(I \otimes U_{\text{phys}})|\psi\rangle = (L^T \otimes I)|\phi\rangle.$$

Hence

$$((L^T)^{-1} \otimes U_{\text{phys}})|\psi\rangle = |\phi\rangle.$$

Theorem 1.1 makes every displayed local factor Clifford. Entrywise conjugation preserves the finite-field Weyl group, since $\overline{X(a)} = X(a)$ and $\overline{Z(b)} = Z(-b)$; it therefore preserves the Clifford normalizer. Taking adjoints preserves the normalizer as well, and $U^T = (\overline{U})^\dagger$. Thus transposition preserves the finite-field Clifford group, and L is Clifford. $\square$

5 Discreteness and quantitative stability

The rigidity theorem uses the stabilizer hypothesis to identify the symmetry group. Finiteness of that group needs far less: two-party marginals alone, at any local dimension, with no stabilizer structure. Throughout this subsection $q \geq 2$ is an arbitrary integer; only the statements naming Clifford operators return to $q = p^e$.

Call a pure state $|\psi\rangle \in (\mathbb{C}^q)^{\otimes n}$ *2-uniform* if

$$\rho_{\{j,k\}}^\psi = q^{-2} I \quad \text{for all } j \neq k.$$

Tracing once more makes every one-party marginal $q^{-1} I$. Such states exist only for $n \geq 4$, since at $n = 3$ purity forces $\text{rank } \rho_{\{1,2\}} = \text{rank } \rho_{\{3\}} \leq q$; and every $\text{AME}(2m, q)$ state with $m \geq 2$ is 2-uniform, stabilizer or not. Write

$$G(\psi) = \{U_1 \otimes \cdots \otimes U_n : (U_1 \otimes \cdots \otimes U_n)|\psi\rangle \in \mathbb{C}|\psi\rangle\}$$

for its product-unitary symmetry group and, for a product unitary U ,

$$\varepsilon(U) = \min_\theta \|U|\psi\rangle - e^{i\theta}|\psi\rangle\| = (2 - 2|\langle\psi|U|\psi\rangle|)^{1/2}$$

for its defect. Because operators on distinct sites commute, a product unitary is a single exponential: $\bigotimes_j e^{iH_j} = e^{iL}$ with $L = \sum_j H_j^{(j)}$, where $H^{(j)}$ denotes H acting at site j . Splitting each H_j into a traceless part h_j and a multiple of the identity leaves $L = M + cI$ with $M = \sum_j h_j^{(j)}$. Everything below rests on one computation about M .

Figure 3 gives a roadmap for the local estimates, the Clifford separation step, and the resulting explicit threshold.

Lemma 5.1 (Local-generator isometry). Let $|\psi\rangle$ be 2-uniform and let h_j and h'_j , $1 \leq j \leq n$, be traceless Hermitian operators on $\mathbb{C}^q$. Put $M = \sum_j h_j^{(j)}$ and $M' = \sum_j h'_j{}^{(j)}$. Then $\langle\psi|M|\psi\rangle = 0$ and

$$\langle M|\psi\rangle, M'|\psi\rangle\rangle = \frac{1}{q} \sum_{j=1}^n \text{Tr}(h_j h'_j). \quad (5.1)$$

In particular $\|M|\psi\rangle\|^2 = q^{-1} \sum_j \|h_j\|_F^2$, so $(h_1, \dots, h_n) \mapsto \sqrt{q} M|\psi\rangle$ is an isometry from the traceless local generators, carrying the product Frobenius form, into the Hilbert space regarded as a real inner product space.

Proof. The one-party marginals give $\langle M \rangle = \sum_j \text{Tr}(\rho_{\{j\}}^\psi h_j) = \sum_j \text{Tr}(h_j)/q = 0$. Expand $\langle\psi|MM'|\psi\rangle$. The terms with equal site indices contribute $\text{Tr}(\rho_{\{j\}}^\psi h_j h'_j) = \text{Tr}(h_j h'_j)/q$. A term

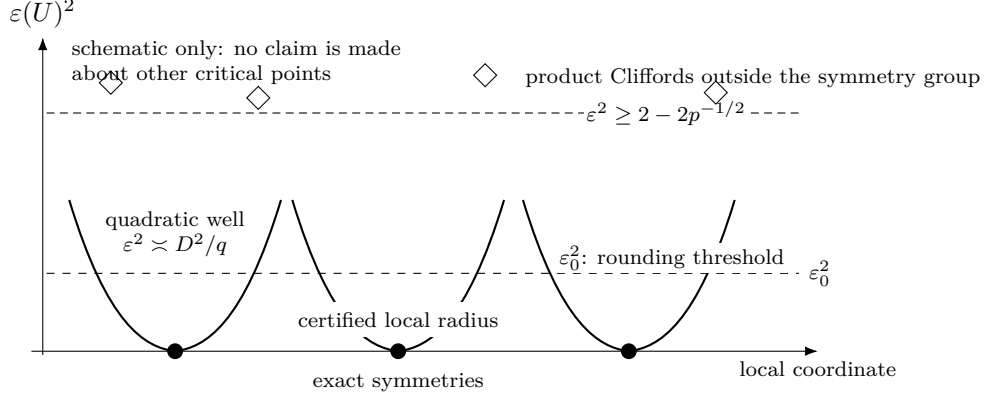


Figure 3: Roadmap for the defect estimates (a one-dimensional schematic of a high-dimensional product-unitary group). Exact symmetries are isolated zeros with certified local quadratic growth. Every product Clifford outside the symmetry group has defect at least $(2 - 2p^{-1/2})^{1/2}$; below the explicit decomposition threshold, an arbitrary product unitary rounds to a Clifford and then to an exact symmetry. The picture asserts neither that every local minimum is Clifford nor that the displayed coordinate exhausts the group.

with $j \neq k$ contributes $\text{Tr}(\rho_{\{j,k\}}^\psi(h_j \otimes h'_k)) = \text{Tr}(h_j) \text{Tr}(h'_k)/q^2 = 0$; both maximal mixedness of the pair marginal and tracelessness are used. Hence $\langle \psi | MM' | \psi \rangle = q^{-1} \sum_j \text{Tr}(h_j h'_j)$, which is real. Since M and M' are Hermitian, the imaginary part of $\langle M\psi, M'\psi \rangle$ is $\langle [M, M'] \rangle / 2i$, and it vanishes as well: operators at distinct sites commute, and the surviving one-site commutators are traceless, so each term is of the form $\text{Tr}(\rho_{\{j\}}^\psi[h_j, h'_j]) = \text{Tr}([h_j, h'_j])/q = 0$. A real symmetric bilinear form agreeing with the product Frobenius form is that form. $\square$

The identity has no error term, and that is the content of the 2-uniformity hypothesis rather than a coincidence: the cross terms cancel exactly because the pair marginals are exactly $q^{-2}I$. A Bell pair is the boundary case. There $\rho_{\{1,2\}}$ is a pure state, the cross term survives, and $M = h \otimes I - I \otimes h^T$ annihilates the state for every h — the $U \otimes \bar{U}$ gauge line that makes $m \geq 2$ sharp in Theorem 1.1.

The second ingredient is that a one-parameter group of product unitaries has a generator of summed local form. This is not automatic: a one-parameter subgroup of $G(\psi)$ is a priori only a continuous homomorphism into $U(q^n)$, and the factors of a product unitary are determined only up to phases, so differentiating factor by factor carries a lifting obligation.

Lemma 5.2 (Product generators). Let $\pi : U(q)^n \rightarrow U(q^n)$ send $(U_1, \dots, U_n)$ to $U_1 \otimes \dots \otimes U_n$. Then π is a smooth homomorphism of compact Lie groups, its image P is a closed subgroup of $U(q^n)$, and

$$\text{Lie}(P) = \left\{ i \sum_j H_j^{(j)} : H_j = H_j^\dagger \text{ on } \mathbb{C}^q \right\}. \quad (5.2)$$

Hence every one-parameter subgroup of a closed subgroup of P has a generator of that form.

Proof. The tensor product of unitaries is multiplicative factor by factor, so π is a smooth homomorphism. Its source is compact, so P is compact and therefore closed, and Cartan's theorem makes P an embedded Lie subgroup of $U(q^n)$. A surjective homomorphism of Lie groups has surjective differential, so $\text{Lie}(P) = d\pi(\mathfrak{u}(q)^n)$, which is the displayed space; its kernel is $\{(i\tau_1 I, \dots, i\tau_n I) : \sum_j \tau_j = 0\}$. A closed subgroup $H \leq P$ is a Lie subgroup with $\text{Lie}(H) \subseteq \text{Lie}(P)$, and the generator of a one-parameter subgroup of H lies in $\text{Lie}(H)$. $\square$

Theorem 5.3 (Discreteness from 2-uniformity). Let $|\psi\rangle$ be a 2-uniform pure state of n qudits of local dimension q . Then $\text{Lie}(G(\psi)) = i\mathbb{R}I$, and $G(\psi)/U(1) \cdot I$ is finite. In particular every $\text{AME}(2m, q)$ state with $m \geq 2$, stabilizer or not, has finite product-unitary symmetry group modulo global phase.

Proof. The set $\{U \in U(q^n) : U|\psi\rangle \in \mathbb{C}|\psi\rangle\}$ is a closed subgroup, so $G(\psi) = P \cap \{\dots\}$ is a closed subgroup of P and hence a compact Lie group. Let $X \in \text{Lie}(G(\psi))$. By Lemma 5.2, $X = iL$ with $L = \sum_j H_j^{(j)}$ Hermitian; write $L = M + cI$ as above. Each e^{itL} fixes the ray of $|\psi\rangle$, so $e^{itL}|\psi\rangle = \lambda(t)|\psi\rangle$ with $\lambda(t) = \langle\psi|e^{itL}|\psi\rangle$ unimodular and smooth in t . Differentiating at $t = 0$ shows that $|\psi\rangle$ is an eigenvector of the Hermitian operator L , say $L|\psi\rangle = \mu|\psi\rangle$ with μ real. Pairing with $|\psi\rangle$ and using $\langle M \rangle = 0$ gives $\mu = c$, hence $M|\psi\rangle = 0$. By Lemma 5.1, $\sum_j \|h_j\|_F^2 = 0$, so every $h_j = 0$ and every H_j is scalar. Thus $\text{Lie}(G(\psi)) = i\mathbb{R}I$. Since $U(1) \cdot I$ is a closed central one-dimensional subgroup of $G(\psi)$, the compact Lie group $G(\psi)/U(1) \cdot I$ has dimension zero and is therefore finite. $\square$

Remark 5.4 (Which torus is divided out). Upstairs in $U(q)^n$ the preimage $\pi^{-1}(G(\psi))$ always contains the local-phase torus $T = U(1)^n$, of dimension n . This does not contradict the theorem: $\pi(T) = U(1) \cdot I$ and $\ker \pi \subset T$ has dimension $n-1$, so $\pi^{-1}(G(\psi))/T \cong G(\psi)/U(1) \cdot I$. Statements about local factors below are statements on $\text{PU}(q)^n$, not statements modulo one global phase.

Remark 5.5 (Discreteness in the literature). Finiteness of this kind is known in two special cases. Wirthmüller determines the identity component of the local-unitary automorphism group of a binary stabilizer code and gives a protected-qubit criterion for finiteness of the induced group [Wir11, Theorem 7 and Corollary 11]; his exclusion of Bell factors is the $n \geq 4$ boundary above. Englbrecht and Kraus determine the local symmetries of qubit stabilizer states, including where continuous ones occur [EK20]. Tan computes the local symmetry group of the four-qutrit AME state outright, of order 5832 [Tan26, Theorem 5.3]. Theorem 5.3 covers arbitrary local dimension and arbitrary, in particular nonstabilizer, 2-uniform states; to our knowledge no statement at that scope is available.

The hypothesis is exactly 2-uniformity, and one-party marginals are not enough. Słowiak, Sawicki, and Maciążek construct locally maximally entangled states — every one-party marginal maximally mixed — whose local-unitary symmetry group contains a nontrivial continuous image of $\text{SU}(N)$ and is therefore positive-dimensional, with dimension growing in the number of parties [SSM21, Lemma 2]. Passing from one-party to two-party marginals is what forces discreteness. In the language of geometric invariant theory the theorem says that a 2-uniform state is a stable point for $\text{SL}(q, \mathbb{C})^n$ rather than merely a critical one; stabilizer dimensions there are known for generic states and in the extremal qubit case [BRVR18, BLRVR19, LWB08, GKW17], but a genericity statement does not decide a particular state.

Remark 5.6 (Two-uniformity does not identify the group). An explicit state shows that the second half of the rigidity phenomenon needs strictly more than the first. Let $q = 2$ and $C = \text{RM}(1, 4) \leq \mathbb{F}_2^{16}$, whose nonzero codeword weights are 8 and 16 and whose dual $\text{RM}(2, 4)$ has minimum distance 4. The equal-phase CSS state of C has label group $C_X \oplus C_Z^\perp$, so by (2.2) a marginal fails to be maximally mixed only when a nonzero word of C or of $C^\perp$ is supported inside it; both distances exceed 2, so the state is 2-uniform and Theorem 5.3 makes its symmetry group finite. That group is not Clifford. Writing $T = \text{diag}(1, e^{i\pi/4})$, the product $T^{\otimes 16}$ multiplies $|x\rangle$ by $e^{i\pi \text{wt}(x)/4}$, which is 1 at every codeword weight, so $T^{\otimes 16}$ fixes the state exactly; and $TX(1)T^\dagger = 2^{-1/2}(X(1) + iX(1)Z(1))$ is not a Weyl axis. Two-uniformity buys finiteness and nothing else. Identifying the finite group as local Clifford is what the AME hypothesis and Theorem 1.1 supply, and the next corollary is that identification.

Corollary 5.7 (Discrete product-unitary symmetry). Fix a stabilizer AME state $|\psi\rangle$ as in Theorem 1.1, and define the local-factor tuple groups

$$G = \{(U_i) \in U(q)^{2m} : (\bigotimes_i U_i)|\psi\rangle \in \mathbb{C}|\psi\rangle\},$$

$$\tilde{G} = \{((U_i), \pi) \in U(q)^{2m} \rtimes S_{2m} : (\bigotimes_i U_i)\pi|\psi\rangle \in \mathbb{C}|\psi\rangle\},$$

where S_{2m} permutes the local factors. Let $T = (S^1)^{2m}$ be the coordinatewise scalar subgroup and put $\Gamma = G/T$, $\tilde{\Gamma} = \tilde{G}/T$. Then

$$1 \longrightarrow T \longrightarrow G \longrightarrow \Gamma \longrightarrow 1, \quad 1 \longrightarrow T \longrightarrow \tilde{G} \longrightarrow \tilde{\Gamma} \longrightarrow 1$$

are short exact sequences with closed connected kernel and finite discrete quotient; both inclusions of T are continuous. In particular, G and $\tilde{G}$ are finite unions of connected components, each a translate of T . Thus the local-factor automorphism group modulo one-site scalar phases is finite and discrete, and its identity component before quotienting is exactly T . If Π is the subgroup of party permutations realized by $\tilde{G}$, there is a further exact sequence

$$1 \longrightarrow \Gamma \longrightarrow \tilde{\Gamma} \longrightarrow \Pi \longrightarrow 1.$$

This last extension splits exactly when the realized permutations admit a homomorphic choice of projective product-unitary lifts. Its obstruction is a nonabelian factor set, computed in Appendix C.

Proof. Every local factor in a tuple from G or $\tilde{G}$ is Clifford. The projective single-qudit Clifford group over a finite field is finite, as is the party-permutation group. The kernel of projectivizing each local factor is exactly the torus of one-site scalar phases. That torus is the identity component, hence is closed; the finite Hausdorff quotient is discrete. Translation identifies every connected component with the scalar torus, so finitely many quotient representatives cover the group. Forgetting the local factors maps $\tilde{\Gamma}$ onto the realized permutation subgroup Π . Its kernel consists exactly of the fixed-party projective automorphisms, which gives the third sequence. A splitting is, by definition, a group-homomorphic right inverse; arbitrary or generator-by-generator lifts do not provide one. $\square$

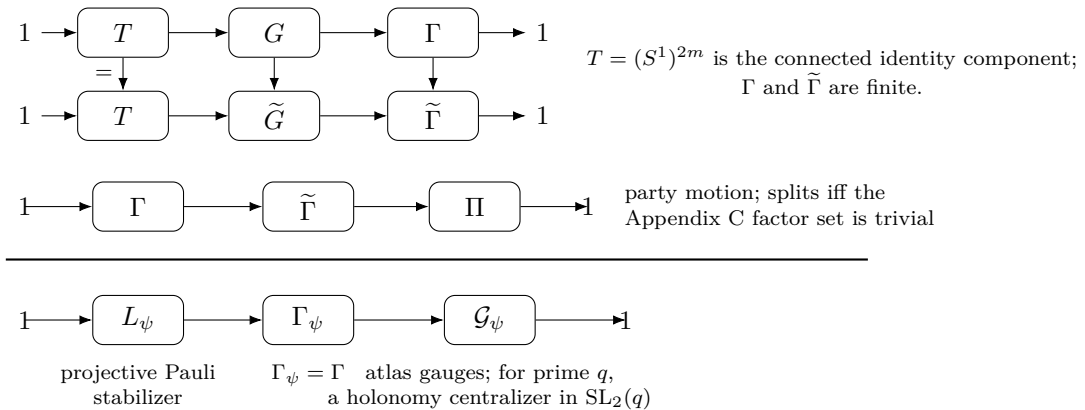


Figure 4: The layered symmetry group of a stabilizer AME state. One-site phases form the identity component, with finite fixed-party and party-moving projective quotients. The fixed-party quotient is itself an extension of the atlas gauge group by the projective Pauli stabilizer. Independently, allowing party motion extends the fixed-party projective group by the realized permutation group, with splitting governed by a nonabelian factor set.

Discreteness makes the exact symmetries isolated. The same identity (5.1) says how fast the defect grows away from them, with a constant that does not involve the number of parties.

Theorem 5.8 (Local stability). Let $|\psi\rangle$ be 2-uniform, let each h_j be traceless Hermitian on $\mathbb{C}^q$, and put

$$U = \bigotimes_{j=1}^n e^{ih_j}, \quad t = \sum_{j=1}^n \|h_j\|_{\text{op}}, \quad D = \left(\sum_{j=1}^n \|h_j\|_F^2 \right)^{1/2}.$$

If $t \leq \frac{1}{2}$ then

$$\frac{D^2}{q} \left(1 - \frac{t}{3} \right) \leq \varepsilon(U)^2 \leq \frac{D^2}{q} \left(1 + \frac{t}{3} \right). \quad (5.3)$$

In particular $D \leq \sqrt{6q/5} \varepsilon(U) < 1.096 \sqrt{q} \varepsilon(U)$, and $\varepsilon(U)^2/D^2 \rightarrow q^{-1}$ as $t \rightarrow 0$, uniformly in n .

Proof. Since the sites commute, $U = e^{iM}$ with $M = \sum_j h_j^{(j)}$, and $\|M\|_{\text{op}} \leq t$ by the triangle inequality. Write $\langle \cdot \rangle = \langle \psi | \cdot | \psi \rangle$. Taylor's formula with integral remainder gives $e^{ix} = 1 + ix - x^2/2 + r(x)$ with $|r(x)| \leq |x|^3/6$ for real x . Let $M = \sum_i \lambda_i P_i$ be the spectral decomposition and put $p_i = \langle P_i \rangle \geq 0$, so that $\sum_i p_i = 1$. Then $\langle r(M) \rangle = \sum_i p_i r(\lambda_i)$ and

$$|\langle r(M) \rangle| \leq \frac{1}{6} \sum_i p_i |\lambda_i|^3 \leq \frac{t \langle M^2 \rangle}{6}.$$

By Lemma 5.1, $\langle M \rangle = 0$ and $\langle M^2 \rangle = D^2/q$, so $\langle e^{iM} \rangle = 1 - \langle M^2 \rangle/2 + \langle r(M) \rangle$. Also $\langle M^2 \rangle \leq t^2 \leq \frac{1}{4}$, so $1 - \langle M^2 \rangle/2 > 0$ and

$$\left| |\langle e^{iM} \rangle| - \left(1 - \frac{1}{2} \langle M^2 \rangle \right) \right| \leq \frac{t \langle M^2 \rangle}{6},$$

using $|\langle e^{iM} \rangle| \geq \text{Re} \langle e^{iM} \rangle$ for the lower estimate. Since $\varepsilon(U)^2 = 2 - 2|\langle e^{iM} \rangle|$, this is exactly (5.3). At $t \leq \frac{1}{2}$ the left inequality reads $\varepsilon^2 \geq \frac{5}{6} D^2/q$, and the two-sided form gives the limit. $\square$

The constant $\sqrt{6/5} = 1.0954\dots$ is therefore sharp to within about ten percent: the true asymptotic constant is $\sqrt{q}$, and the excess is the cubic Taylor cutoff, not structure.

Remark 5.9 (The stability constant is a Fisher metric). Identity (5.1) also explains why the constant does not involve n . For the family $U(s) = \bigotimes_j e^{ish_j} = e^{isM}$, the quantum Fisher information of a pure state is $4 \text{Var}_\psi(M)$ [BC94], so

$$F_Q = \frac{4}{q} \sum_{j=1}^n \|h_j\|_F^2.$$

Splitting the variance of a sum of one-site terms into single-party variances and cross-party covariances is the standard framework for metrologically useful entanglement [HLK⁺12, Tó12]; 2-uniformity is exactly the hypothesis that annihilates every covariance and pins every single-party variance. On traceless local generators — and only there, since (5.1) says nothing about generators that are not sums of one-site terms — the Fisher form is a flat multiple of the Euclidean one. A degenerate direction would be a continuous product symmetry, which Theorem 5.3 excludes; an enhanced direction, of the kind a GHZ state supplies, would need the pair correlations that 2-uniformity forbids. The stability constant is the inverse of that flat metric.

The Fisher form is quadratic data, so it governs the constant and says nothing about how large a neighbourhood the quadratic estimate covers. That size is fixed by the higher moments of M , and Theorem 5.12 and Proposition 5.15 below determine it.

The size of the certified region is controlled by the state's uniformity order, not by the number of parties. Call $|\psi\rangle$ *k-uniform* if every reduced operator on at most k parties is maximally mixed; 2-uniformity is the case $k = 2$, and an $\text{AME}(2m, q)$ state is m -uniform. The hypothesis $t \leq \frac{1}{2}$ in Theorem 5.8 pays for a cubic Taylor remainder. Each additional order of uniformity removes one order from that remainder, because the state and the tracial state assign the same value to every power of M below the uniformity order.

Lemma 5.10 (Tracial agreement below the uniformity order). Let $|\psi\rangle$ be k -uniform on n qudits. Then $\langle\psi|O|\psi\rangle = q^{-n} \text{Tr } O$ for every operator O supported on at most k parties. In particular, with $M = \sum_j h_j^{(j)}$ as above, $\langle\psi|M^r|\psi\rangle = q^{-n} \text{Tr}(M^r)$ for every $r \leq k$.

Proof. If $O = O_S \otimes I$ with $|S| \leq k$ then $\langle O \rangle = \text{Tr}(\rho_S^\psi O_S) = q^{-|S|} \text{Tr } O_S = q^{-n} \text{Tr } O$. Expanding M^r gives a sum of products of one-site operators touching at most $r \leq k$ sites, so every term is of that shape. $\square$

The tracial state $q^{-n} \text{Tr}$ is the product of the one-site tracial states, and the site summands of M commute, so $q^{-n} \text{Tr } f(M) = \mathbb{E} f(X_1 + \dots + X_n)$ with the X_j independent and X_j uniform on the spectrum of h_j . The tracial main term is therefore a product of one-site characteristic functions, and the following contraction estimate controls each factor.

Lemma 5.11 (Per-site contraction). Let h be traceless Hermitian on $\mathbb{C}^q$ with spectral spread $s = \lambda_{\max}(h) - \lambda_{\min}(h)$. Then

$$|q^{-1} \text{Tr } e^{ih}|^2 \leq 1 - c(s) \|h\|_F^2/q, \quad c(s) = \max\left\{1 - \frac{s^2}{12}, \frac{4}{\pi^2} [s \leq \pi]\right\}.$$

Proof. Write $q^{-1} \text{Tr } e^{ih} = \mathbb{E} e^{iX}$ with X uniform on the spectrum. For an independent copy X' , $|\mathbb{E} e^{iX}|^2 = \mathbb{E} \cos(X - X')$ and $|X - X'| \leq s$. From $\cos u \leq 1 - u^2/2 + u^4/24 = 1 - (u^2/2)(1 - u^2/12)$ and $u^2 \leq s^2$, $\mathbb{E} \cos(X - X') \leq 1 - (1 - s^2/12) \text{Var } X$; from $\cos u \leq 1 - 2u^2/\pi^2$ for $|u| \leq \pi$, $\mathbb{E} \cos(X - X') \leq 1 - (4/\pi^2) \text{Var } X$. Tracelessness gives $\text{Var } X = \text{Tr}(h^2)/q = \|h\|_F^2/q$. $\square$

Theorem 5.12 (Stability at uniformity order k). Let $|\psi\rangle$ be k -uniform with $k \geq 2$, let each h_j be traceless Hermitian with spectral spread at most s , and keep t, D, U as in Theorem 5.8. Put

$$c = c(s), \quad \lambda = \frac{2t^{k-1}}{(k+1)!}, \quad \rho = c - \frac{c^2 D^2}{4q} - 2\lambda.$$

Then $\varepsilon(U)^2 \geq \rho D^2/q$; in particular $\rho > 0$ gives $D \leq \sqrt{q/\rho} \varepsilon(U)$.

Proof. Write $e^{ix} = \sum_{r \leq k} (ix)^r/r! + \varrho(x)$ with $|\varrho(x)| \leq |x|^{k+1}/(k+1)!$. By Lemma 5.10 the state and the tracial state agree on every M^r with $r \leq k$, so $\langle e^{iM} \rangle - q^{-n} \text{Tr } e^{iM} = \langle \varrho(M) \rangle - q^{-n} \text{Tr } \varrho(M)$. For any spectral weight vector $p_i \geq 0$ with $\sum_i p_i = 1$,

$$\left| \sum_i p_i \varrho(\lambda_i) \right| \leq \frac{\|M\|_{\text{op}}^{k-1}}{(k+1)!} \sum_i p_i \lambda_i^2,$$

and both states give M^2 the value D^2/q , by Lemma 5.10 with $r = 2$ on one side and by tracelessness on the other. Since $\|M\|_{\text{op}} \leq t$, each term is at most $t^{k-1} D^2/(q(k+1)!)$, so the two expectations differ by at most $\lambda D^2/q$. The sites commute, so $q^{-n} \text{Tr } e^{iM} = \prod_j q^{-1} \text{Tr } e^{ih_j}$, and Lemma 5.11 with $1 - y \leq e^{-y}$ gives

$$|\langle e^{iM} \rangle| \leq \prod_j (1 - c \|h_j\|_F^2/q)^{1/2} + \lambda D^2/q \leq e^{-c D^2/(2q)} + \lambda D^2/q.$$

With $x = c D^2/(2q)$ and $1 - e^{-x} \geq x - x^2/2$, $\varepsilon(U)^2 = 2 - 2|\langle e^{iM} \rangle| \geq 2x - x^2 - 2\lambda D^2/q = \rho D^2/q$. $\square$

Corollary 5.13 (The certified radius grows with the uniformity order). Let $|\psi\rangle$ be k -uniform with $k \geq 2$. If every h_j has spectral spread at most $\frac{1}{2}$, if $D \leq \sqrt{q}/2$, and if

$$t \leq R_k := \left(\frac{(k+1)!}{48} \right)^{1/(k-1)},$$

then $D \leq \sqrt{6q/5} \varepsilon(U)$. One has $\log R_k = \log k - 1 + O(\log k/k)$, so $R_k \sim k/e$.

Proof. Spread at most $\frac{1}{2}$ gives $c \geq 1 - \frac{1}{48} = 0.97917$; with $D^2 \leq q/4$, $c^2 D^2/(4q) \leq c^2/16$, so $c - c^2 D^2/(4q) \geq 0.91924$; and $t^{k-1} \leq (k+1)!/48$ gives $2\lambda \leq \frac{1}{12}$. Hence $\rho \geq 0.83591 > \frac{5}{6}$ and $\sqrt{q/\rho} \leq \sqrt{6q/5}$. The asymptotics are Stirling's formula applied to $\log R_k = (\log(k+1)! - \log 48)/(k-1)$. $\square$

At $k = 2$ this is weaker than Theorem 5.8, which reaches $t \leq \frac{1}{2}$ while $R_2 = \frac{1}{8}$; the loss is the passage through the tracial main term, which the direct argument avoids. The gain begins at $k = 3$, where $R_3 = 1/\sqrt{2}$, and grows from there: $R_4 = 1.357$, $R_{10} = 4.548$, $R_{30} = 12.93$. For a stabilizer $\text{AME}(2m, q)$ state $k = m$, so the certified radius grows linearly in the party count. It cannot grow faster.

Proposition 5.14 (Ceiling on the certified radius). For every $q \geq 2$ and every n there are traceless Hermitian h_j on $\mathbb{C}^q$ with $\sum_j \|h_j\|_{\text{op}} = 2\pi(1 - 1/q)n$ and $D > 0$ for which $\bigotimes_j e^{ih_j}$ is a scalar, hence $\varepsilon = 0$. No estimate $\varepsilon(U)^2 \geq \rho D^2/q$ with $\rho > 0$ can therefore hold on a region of radius exceeding $2\pi(1 - 1/q)n$.

Proof. Take $h_j = (2\pi/q) \text{diag}(q-1, -1, \dots, -1)$ at every site. It is traceless, and both eigenvalues of e^{ih_j} equal $e^{-2\pi i/q}$, so $e^{ih_j} = e^{-2\pi i/q} I$ and the product of n of them is a global phase. Here $\|h_j\|_{\text{op}} = 2\pi(q-1)/q$ and $\|h_j\|_F^2 = (2\pi/q)^2 q(q-1) > 0$. $\square$

Linear growth in the party count is therefore the maximal possible order, and Corollary 5.13 attains it for AME states, within a factor $4\pi e(1 - 1/q)$ of the ceiling. Bounded uniformity order is a different regime, and the following family measures it exactly.

Proposition 5.15 (The certified radius at uniformity order three). Let $\ell \geq 3$, $N = 2^\ell$, and let $|C\rangle$ be the equal-phase CSS state of $C = \text{RM}(1, \ell) \leq \mathbb{F}_2^N$. Its uniformity order is 3, for every ℓ . Take $h_j = cZ(1)$ at every site and put $\theta = Nc \in (0, 2\pi)$. Then

$$\sum_j \|h_j\|_{\text{op}} = \theta, \quad D = \frac{\theta\sqrt{2}}{\sqrt{N}}, \quad \varepsilon = \frac{2|\sin(\theta/2)|}{\sqrt{N}}, \quad \frac{D}{\sqrt{2}\varepsilon} = \frac{\theta}{2\sin(\theta/2)}.$$

The last ratio does not depend on N and increases on $(0, 2\pi)$; it passes $\sqrt{6/5}$ at the unique $\theta^* = 1.46562\dots$. Hence the conclusion of Theorem 5.8 fails at $t = \theta^*$, for every length. At uniformity order three the exact radius for the constant $\sqrt{6q/5}$ therefore lies between $R_3 = 1/\sqrt{2}$ and θ^* .

Proof. The nonzero weights of $\text{RM}(1, \ell)$ are $N/2$ and N , and $C^\perp = \text{RM}(\ell-2, \ell)$ has minimum distance 4. A marginal of the equal-phase CSS state fails to be maximally mixed only when a nonzero word of C or of $C^\perp$ is supported inside it, so by (2.2) the uniformity order is $\min(N/2, 4) - 1 = 3$ whenever $N \geq 8$. Since $Z(1)|x\rangle = (-1)^x|x\rangle$ at a single site, $U|x\rangle = e^{ic(N-2\text{wt}(x))}|x\rangle$, and the weight distribution 1, $2N-2$, 1 on weights 0, $N/2$, N gives $\langle U \rangle = 1 - (1 - \cos \theta)/N$. This is nonnegative, so $\varepsilon^2 = 2 - 2\langle U \rangle = 4\sin^2(\theta/2)/N$. Also $\|cZ(1)\|_F^2 = 2c^2$ and $\|cZ(1)\|_{\text{op}} = c$, giving the displayed D and θ . Finally $\theta/(2\sin(\theta/2)) = u/\sin u$ with $u = \theta/2 \in (0, \pi)$, which increases from 1 to ∞ ; at θ^* the ratio exceeds the constant of Theorem 5.8 while the defect is $2\sin(\theta^*/2)N^{-1/2}$. $\square$

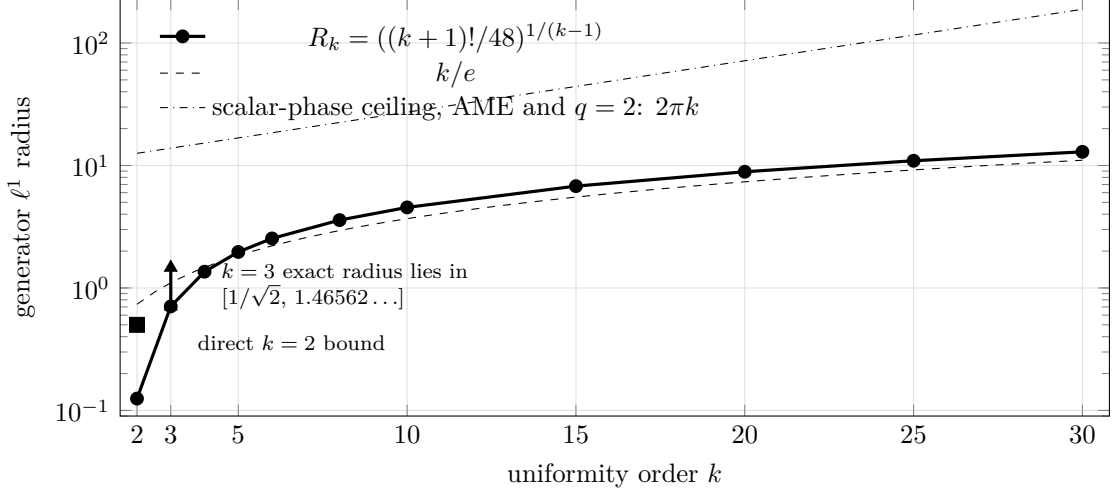


Figure 5: Certified generator ℓ^1 radius of the quadratic stability estimate against uniformity order. The general radius $R_k = ((k+1)!/48)^{1/(k-1)}$ grows as k/e ; scalar-phase configurations bound every such radius by $2\pi(1 - 1/q)n$, so linear growth is the maximal order. For an $\text{AME}(2m, q)$ state, $k = m$. At $k = 3$, the Reed–Muller family pins the exact radius for the constant $\sqrt{6q/5}$ inside the displayed interval.

Two consequences. The smallness hypothesis of Theorem 5.8 is load-bearing, and its constant $\frac{1}{2}$ is within a factor of about three of sharp, since the family leaves the hypothesis at $\theta = \frac{1}{2}$ and violates the conclusion at θ^* . And the party count is not what governs the region. The family has N parties and uniformity order three at every length, so what it bounds is the radius at order three; the defect at which it breaks the conclusion is $1.3379\dots N^{-1/2}$ only because a bounded ℓ^1 radius spread evenly over N sites has defect image of order $N^{-1/2}$. When the uniformity order grows with the party count, as it does for an $\text{AME}(2m, q)$ state, the admissible generator radius grows with it; the independent constraint $D \leq \sqrt{q}/2$ then binds before $t \leq R_m \sim m/e$ does. This enlarges a region already described in generator coordinates. It does not imply that every unitary in a constant-radius defect ball has such coordinates. Both statements concern a known state acted on by product unitaries of bounded generator norm; neither certifies an unknown device.

For stabilizer AME states there is a second route to a stability estimate, which buys the same conclusion with a different hypothesis set. It gives up the global ℓ^1 budget on $\sum_j \|h_j\|_{\text{op}}$ entirely, keeps a per-site condition and the global ℓ^2 condition, and pays only in the constant. The mechanism is a bipartition bound in which the Weyl amplitudes of the two halves are compared directly, so no Taylor remainder appears and no budget is needed to control one.

Proposition 5.16 (Half-splitting bound). Let $|\psi\rangle$ be a stabilizer $\text{AME}(2m, q)$ state with label group L and character χ , let $U = \bigotimes_j U_j$ be a product unitary, and put $a_j(v) = q^{-1} \text{Tr}(W_v^\dagger U_j)$, so that $p_j = |a_j|^2$ is a probability distribution on $\mathbb{F}_q^2$ for each j . For a set A of m parties define $u, v \in \mathbb{R}^L$ by

$$u(\ell) = \prod_{j \in A} |a_j(\ell_j)|, \quad v(\ell) = \prod_{j \notin A} |a_j(\ell_j)|.$$

Then $\|u\| = \|v\| = 1$ and $|\langle \psi | U | \psi \rangle| \leq \langle u, v \rangle$, so $\varepsilon(U)^2 \geq \|u - v\|^2$.

Proof. The coordinate restriction $\pi_A : L \rightarrow (\mathbb{F}_q^2)^A$ has kernel the labels supported on the complementary m -set, which by (4.1) is trivial; both sides have q^{2m} elements, so π_A is a bijection, and likewise for π_{A^c} . Hence $\|u\|^2 = \sum_{x \in (\mathbb{F}_q^2)^A} \prod_{j \in A} p_j(x_j) = 1$, and the same for

v . From (2.2) and $\text{Tr}(W_\ell U) = \prod_j \text{Tr}(W_{\ell_j} U_j)$, taking moduli termwise gives $|\langle \psi | U | \psi \rangle| \leq \sum_{\ell \in L} \prod_j |a_j(\ell_j)| = \langle u, v \rangle$, the relabelling $\ell \mapsto -\ell$ being a bijection of L . For unit vectors $\langle u, v \rangle = 1 - \frac{1}{2} \|u - v\|^2$. $\square$

The AME condition supplies more than the two bijections: it pins down a family of labels of minimum weight that straddles the cut, and those are the labels on which u and v are forced apart.

Lemma 5.17 (Cut-transversal labels). With A as above and $j \notin A$, let $L_j = \{\ell \in L : \text{supp } \ell \subseteq A \cup \{j\}\}$. Then $|L_j| = q^2$, every nonzero $\ell \in L_j$ has support exactly $A \cup \{j\}$, and for every $i \in A \cup \{j\}$ the evaluation $\ell \mapsto \ell_i$ is a bijection $L_j \rightarrow \mathbb{F}_q^2$.

Proof. L_j is the kernel of the projection of L to the coordinates $A^c \setminus \{j\}$, which is surjective because it factors the bijection π_{A^c} ; so $|L_j| = q^{2m}/q^{2(m-1)} = q^2$. A nonzero $\ell \in L_j$ has weight at least $m+1$ by (4.1) and support inside a set of that size. For $i \in A \cup \{j\}$ the kernel of $\ell \mapsto \ell_i$ on L_j consists of labels supported on m parties, hence is trivial, and the two sets have the same cardinality. $\square$

Theorem 5.18 (Stability without a generator budget). Let $|\psi\rangle$ be a stabilizer $\text{AME}(2m, q)$ state with $m \geq 3$, let $U = \bigotimes_j U_j$ be a product unitary, and put $\eta_j = 1 - |q^{-1} \text{Tr } U_j|^2$ and $H = \sum_j \eta_j$. If $H \leq \frac{1}{4}$ then $\varepsilon(U)^2 \geq H/4$. Consequently, if $U = \bigotimes_j e^{ih_j}$ with each h_j traceless Hermitian of spectral spread at most π and $D \leq \sqrt{q}/2$, then

$$\varepsilon(U)^2 \geq \frac{cD^2}{4q}, \quad \text{that is} \quad D \leq 2\sqrt{q/c} \varepsilon(U),$$

with $c = c(s)$ as in Lemma 5.11. No bound on $\sum_j \|h_j\|_{\text{op}}$ is assumed.

Proof. Fix an m -set A and write $\eta = \max_j \eta_j \leq H$. For $j \notin A$ and $w \in \mathbb{F}_q^2$ let $\ell^{j,w}$ be the label of Lemma 5.17 with j -coordinate w . Since $\ell^{j,w}$ vanishes off $A \cup \{j\}$,

$$\sum_{w \neq 0} v(\ell^{j,w})^2 = \eta_j \prod_{i \notin A \cup \{j\}} (1 - \eta_i) \geq \eta_j (1 - H).$$

Every A -coordinate of $\ell^{j,w}$ is nonzero, so $p_i(\ell_i^{j,w}) \leq \eta_i$ for $i \in A$; fixing one $i_0 \in A$ and using injectivity of $w \mapsto \ell_{i_0}^{j,w}$, $\sum_{w \neq 0} u(\ell^{j,w})^2 \leq \prod_{i \in A} \eta_i$. Summing over the m choices of j and using $\prod_{i \in A} \eta_i \leq \eta^{m-1} m^{-1} \sum_{i \in A} \eta_i$ bounds the total by $\eta^{m-1} \sum_{i \in A} \eta_i$. All terms of $\|u - v\|^2$ are nonnegative, and $(x - y)^2 \geq x^2/2 - y^2$, so this family contributes at least $\frac{1}{2}(1 - H) \sum_{j \notin A} \eta_j - \eta^{m-1} \sum_{i \in A} \eta_i$. The mirror family, supported on $A^c \cup \{i\}$ for $i \in A$, is disjoint from it for $m \geq 2$ and contributes the reflected bound. Adding,

$$\varepsilon(U)^2 \geq \|u - v\|^2 \geq \left(\frac{1}{2}(1 - H) - \eta^{m-1} \right) H.$$

At $H \leq \frac{1}{4}$ and $m \geq 3$ one has $\eta^{m-1} \leq H^2 \leq \frac{1}{16}$, so the bracket is at least $\frac{5}{16} \geq \frac{1}{4}$. For the second statement, Lemma 5.11 gives $\eta_j \geq c\|h_j\|_F^2/q$, hence $H \geq cD^2/q$; and $1 - |\mathbb{E} e^{iX}|^2 \leq \frac{1}{2} \mathbb{E}(X - X')^2 = \|h_j\|_F^2/q$ gives $H \leq D^2/q \leq \frac{1}{4}$. $\square$

The two routes are a choice of hypothesis set. Corollary 5.13 assumes per-site spread $\frac{1}{2}$, $D \leq \sqrt{q}/2$, and the ℓ^1 budget $t \leq R_k$, and delivers the constant $\sqrt{6q/5}$. Theorem 5.18 keeps $D \leq \sqrt{q}/2$, relaxes the spread hypothesis to π , deletes the budget, and pays $2\sqrt{q/c}$, which is $\pi\sqrt{q}$ at spread π and $2.02\sqrt{q}$ at spread $\frac{1}{2}$. Its per-site spread condition is not decorative: the family of Proposition 5.14 has e^{ih_j} scalar, so every η_j vanishes and the first bound is empty,

and those h_j have spread 2π , exactly where $c(s) \leq 0$ and the second bound is empty too. The spread condition is the residue of the ceiling.

One configuration is worth recording, because it is the natural candidate for a product unitary that Proposition 5.16 might fail to see. At $q = 2$ take $U_j = H$ at every site, so that each p_j is uniform on $\{X, Z\}$; then $u(\ell)v(\ell) = 2^{-m}$ when every $\ell_j \in \{X, Z\}$ and vanishes otherwise, so $\langle u, v \rangle = 2^{-m} |L \cap \{X, Z\}^{2m}|$. Writing $\{X, Z\} = X + \{0, Y\}$, the set $L \cap \{X, Z\}^{2m}$ is empty or a coset of $L \cap \{0, Y\}^{2m}$, and the latter is a binary linear code of length $2m$ whose nonzero words have weight at least $m+1$. Singleton bounds its dimension by m , with equality only for a binary $[2m, m, m+1]$ MDS code; the binary MDS codes are the trivial ones, which forces $m \leq 1$. Hence $\langle u, v \rangle \leq \frac{1}{2}$ and $\varepsilon(H^{\otimes 2m})^2 \geq 1$ for every stabilizer AME($2m, 2$) state with $m \geq 2$. The bound is not merely nonvacuous there; it is within $\sqrt{2}$ of the largest possible defect. What the half-splitting bound discards is phase information, but the mechanism that detects this configuration is a count of codewords in a structured subset of L , and the MDS constraints force that count down.

Corollary 5.19 (Approximate symmetries decompose). Let $|\psi\rangle$ be a stabilizer AME($2m, q$) state with $m \geq 2$. There is an $\varepsilon_0(\psi) > 0$ such that every product unitary U with $\varepsilon(U) < \varepsilon_0$ factors as

$$U = g \cdot \bigotimes_{j=1}^{2m} e^{ih_j}, \quad g \in G(\psi),$$

with the h_j traceless Hermitian, $\sum_j \|h_j\|_{\text{op}} \leq \frac{1}{2}$, and

$$\left(\sum_j \|h_j\|_F^2 \right)^{1/2} \leq \sqrt{6q/5} \varepsilon(U).$$

By Theorem 1.1 the factor g is an exact local-Clifford symmetry.

The size constraint on the h_j is part of the conclusion and cannot be dropped: the generators of Proposition 5.14 are traceless with $\bigotimes_j e^{ih_j}$ scalar, so the defect vanishes while the generator norm does not.

Proof. Work on the compact manifold $X = \text{PU}(q)^{2m}$, on which $f(U) = \varepsilon(U)^2$ is well defined because the defect depends on U only through $|\langle \psi | U | \psi \rangle|$.

Quadratic growth. Let $B \subseteq X$ be the image of $\{(h_j) \text{ traceless} : \sum_j \|h_j\|_{\text{op}} \leq \frac{1}{2}\}$ under $(h_j) \mapsto [\bigotimes_j e^{ih_j}]$. That map is a diffeomorphism near the origin, of matching dimension $2m(q^2 - 1)$, so B is a neighbourhood of the identity coset, and Theorem 5.8 gives $f \geq \frac{5}{6q} \sum_j \|h_j\|_F^2$ on it. For $g \in G(\psi)$ one has $f(gV) = f(V)$: if $g|\psi\rangle = e^{i\theta}|\psi\rangle$ then $\langle \psi | gV | \psi \rangle = e^{-i\theta} \langle \psi | V | \psi \rangle$ after conjugating the left factor, and f depends only on the modulus. Hence the same estimate holds on the left translate gB , with the generators measured from g , which is the form the factorization $U = g \cdot \bigotimes_j e^{ih_j}$ requires.

Isolation and the threshold. The zero set of f on X is the image Z of $G(\psi)$, finite by Theorem 5.3 and Remark 5.4. So $\bigcup_{g \in Z} gB$ is a neighbourhood of Z , its complement K is compact, and $f > 0$ on K ; put $\varepsilon_0 = (\min_K f)^{1/2} > 0$. If $\varepsilon(U) < \varepsilon_0$ then $U \notin K$, so $U \in gB$ for some $g \in Z$, and the first step applies at that g . $\square$

The stabilizer hypothesis enters this corollary only in the last sentence of its statement. The factorization itself holds for every 2-uniform state, with g an exact product-unitary symmetry; and finiteness of the zero set is a consequence of (5.1) rather than a separate input, since quadratic growth already isolates the zeros of f on a compact manifold. The threshold this proof produces comes from compactness and is not exhibited; Theorem 5.29 below replaces it by a formula, at the cost of the stabilizer hypothesis and of a factor exponential in the party count.

Corollary 5.19 measures how far an approximate symmetry of one state is from an exact one. The intertwiner question between two states is different, and its answer is the tool the explicit threshold needs: it converts a small global defect into a nearby Clifford at each party, with a threshold exhibited in closed form.

Lemma 5.20 (Quantitative diagonal-tensor axes). Let $r \geq 3$, let $E_1, \dots, E_r$ be N -dimensional complex inner product spaces with orthonormal bases $\{e_{ij}\}$, and put

$$T = a \sum_{j=1}^N \lambda_j e_{1j} \otimes \cdots \otimes e_{rj}, \quad a > 0, \quad |\lambda_j| = 1.$$

Let T' have the same form in orthonormal bases $\{e'_{ij}\}$ of spaces E'_i , and let $A_i : E_i \rightarrow E'_i$ be unitaries with

$$\|(A_1 \otimes \cdots \otimes A_r)T - T'\| \leq \eta, \quad \delta = \eta/a < 1/\sqrt{2}.$$

Then in each factor there is a unique permutation π_i with

$$\min_{|\alpha|=1} \|A_i e_{ij} - \alpha e'_{i, \pi_i(j)}\| \leq \sqrt{2} \delta \quad (1 \leq j \leq N). \quad (5.4)$$

Proof. Contract both tensors in the first factor against a target frame vector e'_{1k}^* . The contraction of T' has rank one, and the contraction of $(A_1 \otimes \cdots \otimes A_r)T$ is within η of it. Flattening the latter between its first remaining factor and the last $r-2$ exhibits its singular values as $a|\langle e'_{1k}, A_1 e_{1j} \rangle|$, $1 \leq j \leq N$. By the Eckart–Young theorem the squared mass outside the largest is at most η^2 , hence at most δ^2 after dividing by a . Unitarity of A_1 and completeness of the source basis make the total squared mass one, so the dominant coefficient has square at least $1 - \delta^2 > \frac{1}{2}$. Two orthogonal rows cannot select the same dominant column, and the overlap matrix is square, so the selections form a permutation π_1 . Each matched column also has total squared mass one and contains that coefficient, so its off-axis mass is at most δ^2 , and $2(1 - \sqrt{1 - \delta^2}) \leq 2\delta^2$ gives (5.4). Repeat in every factor. $\square$

Proposition 5.21 (Quantitative intertwiner rigidity). Let $q = p^e$, let $m \geq 2$, and let $|\psi\rangle, |\phi\rangle$ be stabilizer AME($2m, q$) states. Put

$$\tau_p = \min\left\{\frac{1}{3}, \frac{\sin(\pi/p)}{2\sqrt{2}}\right\}, \quad \delta = 2q^{(m+1)/2}\varepsilon.$$

Suppose that, after an allowed party permutation,

$$\min_{|\alpha|=1} \|(U_1 \otimes \cdots \otimes U_{2m})|\psi\rangle - \alpha|\phi\rangle\| \leq \varepsilon$$

and that $\delta < \tau_p$. Then for every party i there is a single-qudit Clifford K_i with

$$q^{-1/2}\|U_i - K_i\|_{\text{HS}} \leq \sqrt{2} \delta = 2\sqrt{2} q^{(m+1)/2} \varepsilon. \quad (5.5)$$

For $m = 3$ and equal-phase CSS states of linear $[6, 3, 4]_q$ MDS codes this is the bound $2\sqrt{2} q^2 \varepsilon$ under $2q^2 \varepsilon < \tau_p$.

Proof. Fix an $(m+1)$ -party set S . By (2.2) and Proposition 3.10 the supported label group $L_\psi(S)$ has q^2 elements and projects bijectively onto $\mathbb{F}_q^2$ at every party of S , so the nonidentity part of the reduced operator is

$$T_S = q^{-(m+1)} \sum_{v \neq 0} \lambda_v W_{1, L_1 v} \otimes \cdots \otimes W_{m+1, L_{m+1} v}, \quad |\lambda_v| = 1,$$

where v runs over the nonzero labels at one fixed party of S and each L_i is the resulting parametrization of $L_\psi(S)$ at party i . Normalizing every Weyl operator by $q^{-1/2}$ puts T_S in the form of Lemma 5.20 with $r = m+1 \geq 3$, $N = q^2 - 1$, and $a = q^{-(m+1)}q^{(m+1)/2} = q^{-(m+1)/2}$. Two unit vectors at phase-optimized distance ε have density matrices at trace distance at most 2ε ; partial trace contracts the trace norm, and the Hilbert–Schmidt norm is no larger, so every $(m+1)$ -party marginal satisfies $\eta \leq 2\varepsilon$ and hence $\eta/a \leq 2q^{(m+1)/2}\varepsilon = \delta$. The lemma then matches Weyl axes within $\kappa = \sqrt{2}\delta$.

It remains to turn approximate axes into an exact Clifford, party by party. Extend the matched label permutation by $\pi(0) = 0$; the identity axis is matched exactly. Conjugating a product of two Weyl operators in the two available orders places the axes labelled $\pi(v+w)$ and $\pi(v) + \pi(w)$ within 3κ . Distinct Weyl axes are at phase-minimized distance $\sqrt{2}$, so $\delta < \frac{1}{3}$ forces $\pi(v+w) = \pi(v) + \pi(w)$; the matching was a permutation, so π is $\mathbb{F}_p$ -linear. Conjugation preserves Weyl commutators exactly, replacing the four factors of a commutator by matched axes costs at most 4κ , and distinct p -th root commutator phases are separated by $2\sin(\pi/p)$, so $\delta < \sin(\pi/p)/(2\sqrt{2})$ forces π to preserve the prime-field symplectic form. By (2.0) it is realized by an exact Clifford K_i .

Finally the phases. Put $V = K_i^\dagger U_i$ and expand $V = \sum_x c_x W_x$ with $\sum_x |c_x|^2 = 1$. Axis closeness gives $|q^{-1} \text{Tr}(W_v^\dagger V W_v V^\dagger)| \geq 1 - \kappa^2/2$ for every v , and character orthogonality turns this into $\sum_x |c_x|^4 \geq (1 - \kappa^2/2)^2$. Hence some $|c_x| \geq 1 - \kappa^2/2$, so V lies within κ of a phase times W_x ; absorbing that displacement into K_i gives (5.5). Every party lies in some $(m+1)$ -set, as in Corollary 3.3. $\square$

The exponent in δ is a feature of the states rather than slack in the estimate. The nonidentity part of an $(m+1)$ -party marginal of an $\text{AME}(2m, q)$ state has Hilbert–Schmidt norm $q^{-(m+1)/2}\sqrt{q^2 - 1}$: the marginal is nearly maximally mixed, and the Weyl signal that exposes the local factors is that small, so a global defect is diluted to that scale before any marginal can read it off.

The closest general framework is the perturbation theory of nearly orthogonally decomposable tensors, which gives dimension-independent bounds without a singular-value gap [AY20]; the argument above is the equal-weight complex specialization needed here, together with the Weyl multiplication, commutator, and implementing-Clifford steps. Its conclusion is distance from the ambient Clifford group and nothing finer. Whenever two states admit an exact intertwiner outside a proposed smaller subgroup, that intertwiner has positive distance from the finite closed subgroup, so taking $\varepsilon = 0$ falsifies every uniform estimate toward it.

Rounding a product unitary to a nearby product Clifford is only half of a decomposition. The other half is deciding whether that Clifford is an exact symmetry, and for stabilizer states it needs no approximation argument at all, because their overlaps take only finitely many values.

Lemma 5.22 (Quantized stabilizer overlap). Let $|\psi\rangle, |\phi\rangle$ be stabilizer states on n qudits of local dimension $q = p^e$, with label groups L_ψ, L_ϕ and stabilizer characters χ_ψ, χ_ϕ . Then

$$|\langle\psi|\phi\rangle|^2 = \begin{cases} |L_\psi \cap L_\phi|/q^n, & \text{if } \chi_\psi = \chi_\phi \text{ on } L_\psi \cap L_\phi, \\ 0, & \text{otherwise.} \end{cases}$$

In particular $|\langle\psi|\phi\rangle|^2 \in \{0\} \cup \{p^{-j} : j \geq 0\}$, so $|\langle\psi|\phi\rangle| > p^{-1/2}$ forces $|\phi\rangle \in \mathbb{C}|\psi\rangle$.

Proof. By (2.2), $\rho_\psi = q^{-n} \sum_{\ell \in L_\psi} \chi_\psi(\ell) W_\ell$ and likewise for ϕ . Weyl orthogonality $\text{Tr}(W_\ell^\dagger W_{\ell'}) = q^n \delta_{\ell, \ell'}$ gives

$$|\langle\psi|\phi\rangle|^2 = \text{Tr}(\rho_\psi \rho_\phi) = q^{-n} \sum_{\ell \in L_\psi \cap L_\phi} (\chi_\psi \overline{\chi_\phi})(\ell).$$

The summand is a character of the group $L_\psi \cap L_\phi$, so the sum is $|L_\psi \cap L_\phi|$ when that character is trivial and 0 otherwise. The group has p -power order and $q^n = p^{en}$, which gives the quantization; the value 1 occurs only when $L_\psi = L_\phi$ and the characters agree, that is, when the two states span the same ray. $\square$

The quantization has a consequence about the symmetry group that is stronger, and easier to remember, than any threshold formula. Theorem 5.3 makes $G(\psi)$ finite, hence its elements isolated; on the Clifford locus they are isolated by a fixed amount that no parameter of the problem can erode.

Corollary 5.23 (Uniform separation of the exact symmetries). Let $|\psi\rangle$ be a stabilizer state on n qudits of local dimension $q = p^e$, and let K be a product Clifford. Then either $K \in G(\psi)$ or

$$\varepsilon(K) \geq (2 - 2p^{-1/2})^{1/2} \geq (2 - \sqrt{2})^{1/2} = 0.7653\dots$$

Equivalently, if K and K' are product Cliffords in different cosets of $G(\psi)$, then $K^\dagger K'$ has defect at least this much. The bound depends on the characteristic alone: not on the number of parties, not on the state, and not on the local dimension beyond its characteristic.

Proof. $K|\psi\rangle$ is a stabilizer state, so Lemma 5.22 applies to the pair $|\psi\rangle, K|\psi\rangle$: either $|\langle\psi|K|\psi\rangle| \leq p^{-1/2}$, whence $\varepsilon(K)^2 = 2 - 2|\langle\psi|K|\psi\rangle| \geq 2 - 2p^{-1/2}$, or the overlap exceeds $p^{-1/2}$ and $K|\psi\rangle \in \mathbb{C}|\psi\rangle$. For two product Cliffords apply this to $K^\dagger K'$, which is again one. Since $p \geq 2$, the bound is least at $p = 2$. $\square$

So on the Clifford locus the defect has no small nonzero values. This is what removes the compactness step from Corollary 5.19: a product unitary with small defect is first rounded to a product Clifford by Proposition 5.21, and the rounded operator is then an exact symmetry outright, because a near-miss is impossible.

The rounding need not inspect one minimum-support marginal in isolation. All such marginals are projections of orthogonal exact-support sectors of the same global density-matrix difference. Summing their squared errors therefore recovers part of the signal lost by marginalization.

Lemma 5.24 (Collective minimum-support energy). Let $\rho = |\psi\rangle\langle\psi|$, $\Delta = U\rho U^\dagger - \rho$, $n = 2m$, and $r = m + 1$. Then

$$\sum_{\substack{A \subseteq [n] \\ |A|=r}} q^r \|\Delta_A\|_{\text{HS}}^2 \leq q^n \|\Delta\|_{\text{HS}}^2 \leq 2q^n \varepsilon(U)^2. \quad (5.7)$$

Writing $\delta_A = q^{r/2} \|\Delta_A\|_{\text{HS}}$ and $N_m = \binom{2m}{m+1}$, there are two r -sets A, B with $A \cup B = [n]$ and

$$\delta_A^2 + \delta_B^2 \leq \frac{4q^n}{N_m} \varepsilon(U)^2. \quad (5.8)$$

Proof. The stabilizer expansion of ρ has no nonidentity component on at most m sites. Local conjugation preserves exact support because it preserves tracelessness. Thus Δ_A , for $|A| = r$, is precisely the partial trace of the exact-support- A component of Δ . If that component is $D_A \otimes I_{A^c}$, then $\Delta_A = q^{n-r} D_A$ and

$$\|D_A \otimes I_{A^c}\|_{\text{HS}}^2 = q^{-(n-r)} \|\Delta_A\|_{\text{HS}}^2.$$

Exact-support subspaces are mutually orthogonal, which gives the first inequality in (5.7). The second follows from $\|\Delta\|_{\text{HS}}^2 = 2(1 - |\langle\psi|U|\psi\rangle|^2) \leq 2\varepsilon(U)^2$.

Join two r -sets when their union is $[n]$. This graph is regular: for fixed A , the complement of a neighbor is any $(m-1)$ -subset of A . Averaging $\delta_A^2 + \delta_B^2$ over its edges gives twice the vertex average. One edge therefore satisfies (5.8). $\square$

The two good sets cover every party, so the quantitative-axis argument can be applied to A and B , using A on their intersection. This gives one product Clifford while retaining collective ℓ^2 control of the local errors. Two-uniformity then converts that control directly to state distance, without a telescoping ℓ^1 estimate.

Theorem 5.25 (Aggregate global rounding). Let $|\psi\rangle$ be a stabilizer AME($2m, q$) state with $q = p^e$ and $m \geq 3$. Put

$$N_m = \binom{2m}{m+1}, \quad r = m+1, \quad d_p = (2 - 2p^{-1/2})^{1/2},$$

$$R_{m,q} = \min \left\{ \frac{\tau_p \sqrt{N_m}}{2q^m}, \frac{\sqrt{N_m}}{2q^{m+1/2}}, \frac{\sqrt{N_m}}{2\pi\sqrt{2r}q^m}, \frac{d_p}{1 + \pi\sqrt{2rq^{2m}/N_m}} \right\}. \quad (5.9)$$

If $\varepsilon(U) < R_{m,q}$, then, up to a global phase,

$$U = g \bigotimes_{i=1}^{2m} e^{ih_i}, \quad g \in G(\psi),$$

where the h_i are traceless Hermitian, have spectral spread at most π , and satisfy

$$\left(\sum_i \|h_i\|_F^2 \right)^{1/2} \leq \pi\sqrt{q}\varepsilon(U). \quad (5.10)$$

Moreover $R_{m,q} = \Theta_q(m^{-3/4}(2/q)^m)$. Thus this route is polynomial in m at $q = 2$, whenever the stated AME state exists, improves the exponential base at $q = 3$, improves the frame-rounding prefactor at $q = 4$, and is superseded by Theorem 5.29 for $q > 4$.

Proof. Choose A, B from Lemma 5.24. The nonidentity part of each marginal has diagonal Weyl coefficient $q^{-r/2}$, so Lemma 5.20 and its additivity, commutator, and phase steps give a Clifford K_i at each party with

$$q^{-1/2}\|U_i - K_i\|_{\text{HS}} \leq \sqrt{2}\delta_i, \quad \sum_i \delta_i^2 \leq r(\delta_A^2 + \delta_B^2) \leq \frac{4rq^n}{N_m}\varepsilon(U)^2, \quad (5.11)$$

where $\delta_i = \delta_A$ on A and $\delta_i = \delta_B$ on $A^c \subset B$. The first clause of (5.9) supplies the axis threshold τ_p .

Choose principal eigenangles of $K_i^\dagger U_i$ after optimizing its scalar phase and subtract their mean, obtaining traceless h_i . Since $|x| \leq (\pi/2)|e^{ix} - 1|$ for $|x| \leq \pi$,

$$D^2 := \sum_i \|h_i\|_F^2 \leq \frac{\pi^2 q}{2} \sum_i \delta_i^2 \leq \frac{2\pi^2 q r q^n}{N_m} \varepsilon(U)^2. \quad (5.12)$$

The second clause of (5.9) puts all eigenangles in an arc of length π ; the third gives $D \leq \sqrt{q}/2$.

Put $K = \bigotimes_i K_i$ and $V = K^\dagger U = \bigotimes_i e^{ih_i}$, up to phase. If $M = \sum_i h_i^{(i)}$, two-uniformity and tracelessness give $\langle M^2 \rangle = D^2/q$. Hence

$$\|(V - I)\psi\|^2 = \langle 2 - 2\cos M \rangle \leq \langle M^2 \rangle = D^2/q.$$

Together with (5.12), the fourth clause of (5.9) puts the rays of $K\psi$ and ψ less than d_p apart. Corollary 5.23 forces $K \in G(\psi)$. Apply Theorem 5.18 to the residual V : its constant has $c \geq 4/\pi^2$, giving (5.10).

Finally $N_m \sim 4^m/\sqrt{\pi m}$. The third and fourth clauses of (5.9) have order $m^{-3/4}(2/q)^m$, and the other two are larger by a polynomial factor. Comparison with the $q^{-(m+2)/2}/m$ scale below gives the stated boundary at $q = 4$. $\square$

The theorem separates frame recognition from the choice of residual metric. If the ℓ^1 conclusion of Corollary 5.19 is required, the same covering pair and a telescoping estimate give it below

$$\min \left\{ \frac{\tau_p \sqrt{N_m}}{2q^m}, \frac{\sqrt{N_m}}{4\pi\sqrt{2q}\sqrt{(m+1)^2 + (m-1)^2} q^m} \right\}. \quad (5.13)$$

At $q = 2$ this is of order $m^{-5/4}$. The loss relative to (5.9) comes from imposing a global generator ℓ^1 budget after the frames are known; it is not a remaining marginal-dilution loss.

There is a stronger global route which does not estimate support sectors at all. Choose one AME party as a logical input. The other $2m - 1$ parties form an $[[2m - 1, 1, m]]_q$ stabilizer code, and the product of their local factors approximately implements the transpose-inverse of the factor on the chosen party. Three correctable regions then test that logical action by a nested commutator. The exact mechanism is standard: the stabilizer cleaning lemma is due to Bravyi and Terhal [BT09, Lemma 1], nested-commutator localization underlies the Bravyi–König hierarchy theorem [BK13], and Pastawski–Yoshida prove directly that a transversal logical unitary on three cleanable regions is Clifford [PY15, Lemma 5]. Disjointness gives a complementary exact-code hierarchy bound [JOKY18, Theorem 5]. The point below is therefore not a new qualitative transversal-gate obstruction. It is the robust step: an implementation with leakage error ε forces every nested logical Weyl commutator within 8ε of a scalar, after which a Fourier argument gives an explicit Clifford distance.

For an encoding isometry $V : \mathbb{C}^q \rightarrow \mathcal{H}$, write

$$E(T, L) = q^{-1/2} \|TV - VL\|_{\text{HS}}.$$

It satisfies $E(TS, LM) \leq E(T, L) + E(S, M)$ and $E(T^\dagger, L^\dagger) = E(T, L)$.

Lemma 5.26 (Quantitative cleaning commutator). Let V encode one logical qudit in a stabilizer code whose physical parties have a partition into three correctable regions. Let T be transversal, and suppose $E(T, L) \leq \varepsilon$. For all logical Weyl operators P, Q ,

$$q^{-1/2} \min_{|z|=1} \| [L, P], Q - zI \|_{\text{HS}} \leq 8\varepsilon, \quad (5.14)$$

where $[A, B] = ABA^\dagger B^\dagger$.

Proof. Let R_1, R_2, R_3 be the assumed correctable partition. The stabilizer cleaning lemma supplies a representative $\tilde{P}$ of P outside R_1 and a representative $\tilde{Q}$ of Q outside R_2 . For reference, cleaning here is the elementary symplectic statement that the restriction of a normalizer label to a correctable region lies in the restriction of the stabilizer label space; subtracting that stabilizer removes the restriction.

Transversality makes $[T, \tilde{P}]$ supported outside R_1 , so $C = [[T, \tilde{P}], \tilde{Q}]$ is supported on $R_1^c \cap R_2^c = R_3$. The composition rules give

$$E(C, [[L, P], Q]) \leq 4\varepsilon. \quad (5.15)$$

Knill–Laflamme correction on R_3 gives $V^\dagger C V = cI$. Hence the logical nested commutator lies within 4ε of cI . Its normalized Hilbert–Schmidt norm is one, so $|c| \geq 1 - 4\varepsilon$; replacing c by its phase proves (5.14). Notice that C need not preserve the code. Only its scalar compression is used, while (5.15) controls leakage. $\square$

The remaining step is finite-dimensional Fourier concentration. It is stated separately because it is the point where scalar commutators become Clifford axes.

Lemma 5.27 (Nested Weyl commutators round to a Clifford). Let $L \in U(q)$, $q = p^e$, and suppose for all Weyl labels v, w that

$$q^{-1/2} \min_{|z|=1} \| [L, W_v], W_w - zI \|_{\text{HS}} \leq \eta.$$

If

$$\eta < \tau_p := \min \left\{ \frac{1}{3}, \frac{\sin(\pi/p)}{2\sqrt{2}} \right\},$$

then some additive one-qudit Clifford K satisfies

$$q^{-1/2} \min_{|z|=1} \| L - zK \|_{\text{HS}} \leq \eta.$$

Proof. Put $A_v = LW_vL^\dagger$. Conjugation of W_w by W_v changes only its phase, so every $[A_v, W_w]$ is within η of a scalar. We use the following Fourier observation twice. If $A = \sum_a c_a W_a$ is unitary and every $[A, W_w]$ is within η of a scalar, then A lies within η of one Weyl axis. Indeed, with $p_a = |c_a|^2$, the commutator estimate gives

$$\left| \sum_a p_a \chi_w(a) \right| \geq 1 - \eta^2/2.$$

Character orthogonality, averaged over w , yields $\sum_a p_a^2 \geq (1 - \eta^2/2)^2$. Thus $\max_a p_a \geq (1 - \eta^2/2)^2$, which is precisely the asserted normalized distance bound.

Apply the observation to each A_v , and call its unique nearby Weyl label $F(v)$. The exact identity $A_{v+u} \in \mathbb{C}A_vA_u$ puts the axes labelled by $F(v+u)$ and $F(v)+F(u)$ within 3η . Distinct Weyl axes have phase-optimized distance $\sqrt{2}$, so $\eta < 1/3$ makes F additive. Comparing the exact commutator phases of A_v, A_u with those of their nearby Weyls costs at most 4η . The least nontrivial p -th-root chord is $2\sin(\pi/p)$, so the stated bound makes F trace-symplectic.

Choose an additive Clifford K_0 inducing F . Then $K_0^\dagger L$ approximately fixes every Weyl axis, so all its Weyl commutators are within η of scalars. The Fourier observation puts it within η of a Weyl; multiplying that Weyl into K_0 gives the required Clifford. $\square$

At $\varepsilon = 0$, the preceding two lemmas recover the three-cleanable-region Clifford conclusion of [PY15, Lemma 5]. Their quantitative combination, and the constants in (5.14), are what will be used here.

Theorem 5.28 (Cleaning-based global rounding). Let $|\psi\rangle$ be a stabilizer AME($2m, q$) state, $q = p^e$, $m \geq 2$, and $n = 2m$. Put

$$d_p = (2 - 2p^{-1/2})^{1/2},$$

and

$$R_{n,q}^{\text{clean}} = \min \left\{ \frac{\tau_p}{8}, \frac{1}{4\sqrt{2q}}, \frac{1}{8\pi\sqrt{n}}, \frac{d_p}{1 + 4\pi\sqrt{n}} \right\}. \quad (5.16)$$

If $\varepsilon(U) < R_{n,q}^{\text{clean}}$, then, up to a global phase,

$$U = g \bigotimes_{i=1}^n e^{ih_i}, \quad g \in G(\psi),$$

where the h_i are traceless Hermitian, have spectral spread at most π , and satisfy

$$\left(\sum_i \|h_i\|_F^2 \right)^{1/2} \leq \pi\sqrt{q}\varepsilon(U). \quad (5.17)$$

Before the exact-symmetry step, the recovered local Cliffords obey

$$q^{-1/2} \min_{|z|=1} \|U_i - zK_i\|_{\text{HS}} \leq 8\varepsilon(U) \quad (5.18)$$

at every party. For fixed q , (5.16) has order $n^{-1/2}$, conditional on existence of the stated AME states.

Proof. Choose party i as the logical leg and write the AME Choi form as $q^{-1/2} \sum_a |a\rangle \otimes V_i |a\rangle$. The other local factors form a transversal physical unitary T_i , and vectorization gives a logical unitary L_i , obtained from U_i by transpose and inverse, with $E(T_i, L_i) = \varepsilon(U)$. Lemmas 5.26 and 5.27, with $\eta = 8\varepsilon(U)$, give (5.18); transpose and inverse preserve the additive Clifford group.

Put $K = \bigotimes_i K_i$, choose principal logarithms of $K_i^\dagger U_i$ after optimizing the local phases, and let $D^2 = \sum_i \|h_i\|_F^2$. The chord-angle bound gives

$$D^2 \leq \frac{\pi^2 q}{4} \sum_i \left(q^{-1/2} \|U_i - K_i\|_{\text{HS}} \right)^2 \leq 16\pi^2 q n \varepsilon(U)^2. \quad (5.19)$$

The second clause of (5.16) puts every spectrum in an arc of length π ; the third gives $D \leq \sqrt{q}/2$. Two-uniformity gives $\|(K^\dagger U - I)\psi\| \leq D/\sqrt{q} \leq 4\pi\sqrt{n}\varepsilon(U)$. The fourth clause and the triangle inequality put $K\psi$ less than d_p from ψ , so Corollary 5.23 makes K an exact symmetry. Apply Theorem 5.18 to the residual to obtain (5.17). $\square$

If the sharper local constant and the old global generator budget are needed, the same proof gives the conclusion of Corollary 5.19 under

$$\varepsilon(U) < \min \left\{ \frac{\tau_p}{8}, \frac{1}{16\pi n \sqrt{q}} \right\}. \quad (5.20)$$

Indeed (5.18), the operator-norm bound by Hilbert-Schmidt norm, and the principal logarithm give $\sum_i \|h_i\|_{\text{op}} \leq 8\pi n \sqrt{q} \varepsilon(U) < 1/2$; the sharp local theorem then gives $D \leq \sqrt{6q/5} \varepsilon(U)$.

The cleaning route supersedes both explicit radii above as a bound for fixed q : it is polynomial in the party count and has no boundary at $q = 4$. The earlier theorems remain useful because they expose the exact cost of reading one marginal and of aggregating orthogonal minimum-support sectors. The new $n^{-1/2}$ scale is a proved lower bound, not an optimality claim; no product-unitary family is known to saturate the collective frame-error estimate (5.19).

Theorem 5.29 (Explicit decomposition threshold). Let $|\psi\rangle$ be a stabilizer AME($2m, q$) state with $q = p^e$, $m \geq 2$, $n = 2m$. Put

$$\varepsilon_0 = \min \left\{ \frac{\tau_p}{2q^{(m+1)/2}}, \frac{1}{4\sqrt{2}\pi n q^{(m+2)/2}} \right\}, \quad \tau_p = \min \left\{ \frac{1}{3}, \frac{\sin(\pi/p)}{2\sqrt{2}} \right\}.$$

Every product unitary U with $\varepsilon(U) < \varepsilon_0$ satisfies the conclusion of Corollary 5.19.

Proof. Write $\varepsilon = \varepsilon(U)$ and $B = 2\sqrt{2} n q^{(m+2)/2} \varepsilon$, so that $\varepsilon < \varepsilon_0$ gives $B < 1/(2\pi)$.

Rounding. Since $2q^{(m+1)/2} \varepsilon < \tau_p$, Proposition 5.21 applied with $|\phi\rangle = |\psi\rangle$ and the identity permutation supplies Cliffords K_i with $\|U_i - K_i\|_{\text{op}} \leq \|U_i - K_i\|_{\text{HS}} \leq 2\sqrt{2} q^{(m+2)/2} \varepsilon$. Summing over the n parties gives $\sum_i \|U_i - K_i\|_{\text{op}} \leq B$.

The rounded Clifford is exact. Put $K = \bigotimes_i K_i$, so that $\|U - K\|_{\text{op}} \leq B$ and $|\langle \psi | K | \psi \rangle| \geq 1 - \varepsilon^2/2 - B$. Here $B < 1/(2\pi) < 0.1592$, and $\varepsilon \leq \tau_p/(2q^{(m+1)/2}) \leq 1/(6 \cdot 2^{3/2}) < 0.06$

because $q \geq 2$ and $m \geq 2$, so the right side exceeds $1 - 0.0018 - 0.1592 > 1/\sqrt{2} \geq p^{-1/2}$. By Corollary 5.23, $K \in G(\psi)$; set $g = K$.

The residual generators are small. Each factor of $V = g^\dagger U$ satisfies $\min_\varphi \|K_i^\dagger U_i - e^{i\varphi} I\|_{\text{op}} \leq \|U_i - K_i\|_{\text{op}} \leq B < 1$. For a unitary W on $\mathbb{C}^q$ with $\sigma = \min_\varphi \|W - e^{i\varphi} I\|_{\text{op}} \leq 1$, the eigenvalues of $e^{-i\varphi} W$ at the optimal phase are $e^{i\mu_a}$ with $2|\sin(\mu_a/2)| \leq \sigma$ and $\mu_a \in (-\pi, \pi]$, so $|\mu_a| \leq 2 \arcsin(\sigma/2) \leq \pi\sigma/2$; subtracting the mean gives $W = e^{i\varphi'} e^{ih}$ with h traceless Hermitian and $\|h\|_{\text{op}} \leq 2 \max_a |\mu_a| \leq \pi\sigma$. Applying this at each site, $\sum_i \|h_i\|_{\text{op}} \leq \pi B < \frac{1}{2}$, and the collected phases are a global phase absorbed into g . Finally $\varepsilon(V) = \varepsilon(U)$, because g fixes the ray of $|\psi\rangle$, so Theorem 5.8 applies to V on its hypothesis and gives $D \leq \sqrt{6q/5} \varepsilon(U)$. $\square$

Nothing in this proof is a compactness argument: the finite zero set is never enumerated and no minimum over a compact remainder is taken. What compactness supplied in Corollary 5.19 is supplied here by Corollary 5.23, in closed form. The factorization is also constructive in the sense that matters: g is produced from U by rounding each local factor to a Clifford, and an $(m+1)$ -party marginal certifies the rounding.

For this single-marginal route the price is the size of ε_0 , which is $q^{-(n+4)/4}/n$ up to constants and therefore exponentially small in the number of parties. Two steps pay it. Rounding costs $q^{(m+1)/2}$, and by the marginal-dilution remark after Proposition 5.21 no detector that reads a single small marginal can avoid that factor. Summing n per-site operator-norm errors and taking n logarithms costs a further $n\sqrt{q}$, which is the price of a per-site conclusion.

The exponential gap left by this theorem and Theorem 5.25 is closed by Theorem 5.28. The remaining comparison is with Proposition 5.15. That proposition bounds any threshold valid across states of uniformity order three; an $\text{AME}(2m, q)$ state has uniformity order m , and for such states Corollary 5.13 certifies the quadratic estimate on a generator-coordinate region whose ℓ^1 radius grows with m , as does Theorem 5.18 once $m \geq 3$. Those estimates control defect from specified generators; they do not put a defect ball inside that coordinate region. The cleaning theorem supplies the inverse statement at radius of order $n^{-1/2}$, with no claim that this party-count dependence is optimal.

Corollary 5.30 (Local gauge groups of 2-unitary gates). Let W be a two-qudit gate whose associated four-party state $\psi_{ijkl} = q^{-1} W_{kl,ij}$ is $\text{AME}(4, q)$ — equivalently, W is 2-unitary, remaining unitary after realignment and after partial transposition — and let $G(\psi)$ be its product-unitary symmetry group. Then $(u_1, u_2, u_3, u_4) \mapsto (\bar{u}_1, \bar{u}_2, u_3, u_4)$ is an isometric group isomorphism carrying $G(\psi)$ onto the local gauge group

$$\{(v_1 \otimes v_2, u_3 \otimes u_4) : (u_3 \otimes u_4)W(v_1 \otimes v_2)^\dagger = e^{i\varphi} W\},$$

and it preserves $(\sum_j \|h_j\|_F^2)^{1/2}$. Consequently the local gauge group of any 2-unitary gate is finite modulo the local phases, Theorem 5.8 transports with its hypothesis $\sum_j \|h_j\|_{\text{op}} \leq \frac{1}{2}$, and Corollary 5.19 transports with its threshold; for a gate arising from a stabilizer $\text{AME}(4, q)$ state Theorem 5.29 exhibits that threshold at $m = 2$, namely $\varepsilon_0 = \min\{\tau_p/(2q^{3/2}), 1/(16\sqrt{2}\pi q^2)\}$. Gate defects are measured in the metric $q^{-1}\|\cdot\|_F$. For a gate arising from a stabilizer $\text{AME}(4, q)$ state the exact gauge group is local Clifford.

Proof. Normalization is consistent: $\|W\|_F^2 = q^2$ for a unitary on $\mathbb{C}^q \otimes \mathbb{C}^q$, so $\langle \psi | \psi \rangle = \|W\|_F^2/q^2 = 1$. A product unitary acts on the reshaped gate by $W \mapsto (u_3 \otimes u_4)W(u_1 \otimes u_2)^T$, so $\|U|\psi\rangle - e^{i\theta}|\psi\rangle\| = q^{-1}\|(u_3 \otimes u_4)W(u_1 \otimes u_2)^T - e^{i\theta}W\|_F$; this is the stated metric. Transposition preserves the Frobenius norm and $u \mapsto \bar{u}$ is an automorphism of $U(q)$ fixing $\|h\|_F$ for the corresponding generators, and $(u_1 \otimes u_2)^T = (v_1 \otimes v_2)^\dagger$ with $v_a = \bar{u}_a$, which gives the displayed isomorphism. An $\text{AME}(4, q)$ state is 2-uniform, so Theorem 5.3 applies, with Remark 5.4 identifying the quotient as the one by the local phases; the stabilizer case is Theorem 1.1. $\square$

The statement is vacuous at $q = 2$, where no $\text{AME}(4, 2)$ state exists. In the dual-unitary circuit literature the local gauge freedom is the standing nuisance parameter of the classification programme [RAL22], and the finite-field Clifford subfamily has been classified, with perfect-tensor cores appearing among its equivalence classes [Pah26]; the corollary says that for 2-unitary gates the gauge group itself is finite and, in the stabilizer case, Clifford.

6 Exact MDS–CSS logical images and lifts

Proposition 6.1 (The diagonal-multiplier line). Let $E, F \leq \mathbb{F}_q^{2m}$ be linear $[2m, m, m+1]_q$ MDS codes, and put

$$\mathcal{D}(E, F) = \{s \in \mathbb{F}_q^{2m} : \text{diag}(s)E \subseteq F\}.$$

Every nonzero $s \in \mathcal{D}(E, F)$ has full coordinate support and $\text{diag}(s)E = F$. Consequently

$$\dim_{\mathbb{F}_q} \mathcal{D}(E, F) \leq 1, \quad \mathcal{D}(E, E) = \mathbb{F}_q(1, \dots, 1).$$

In particular, $\mathcal{D}(C, C^\perp)$ has dimension zero or one, and C is diagonally isodual exactly in the dimension-one case. When that line exists, any two nonzero witnesses differ by a unique nonzero scalar, so every ratio s_i/s_j is intrinsic.

Proof. Let $s \neq 0$, and write $D = \text{diag}(s)$. If D had support of size at most m , choose a word of E nonzero at a supported coordinate. Its image would be a nonzero word of F of weight at most m , contrary to the distance of F . Thus D has at least $m+1$ nonzero entries. Its kernel on E consists of words supported on at most $m-1$ coordinates, so the distance of E makes that kernel zero. Equal dimensions now give $DE = F$. A zero diagonal entry would force every word of F to vanish at that coordinate, which an MDS code does not, so s has full support.

If two multipliers were independent, a nonzero linear combination could be chosen to vanish at one coordinate, contradicting full support. Hence $\dim \mathcal{D}(E, F) \leq 1$. The all-ones vector belongs to $\mathcal{D}(E, E)$, proving the self-multiplier statement. Taking $F = C^\perp$ gives the remaining assertions. $\square$

Proof of Corollary 1.4. Write a fixed-party product-Clifford action on Pauli labels as

$$F_i = \begin{pmatrix} \alpha_i & \beta_i \\ \gamma_i & \delta_i \end{pmatrix},$$

and let $D_\alpha, D_\beta, D_\gamma, D_\delta$ be the diagonal matrices with the corresponding entries. Preservation of $L_C = C_X \oplus C_Z^\perp$ is equivalent to

$$\begin{aligned} D_\alpha C &\subseteq C, & D_\gamma C &\subseteq C^\perp, \\ D_\beta C^\perp &\subseteq C, & D_\delta C^\perp &\subseteq C^\perp. \end{aligned} \tag{6.1}$$

If the block at the chosen input is nondiagonal, then D_β or D_γ is nonzero. Proposition 6.1 and (6.1) give

$$D_\gamma C = C^\perp \quad \text{or} \quad D_\beta C^\perp = C.$$

The second equality can be inverted, so either one supplies a nonsingular diagonal S with $SC = C^\perp$. Consequently, on the non-isodual branch every input block is diagonal. Its determinant is one, and the common blocks $\text{diag}(a, a^{-1})$ always preserve L_C ; hence the exact linear factor there is T .

Now suppose

$$S = \text{diag}(s_1, \dots, s_{2m}), \quad SC = C^\perp. \tag{6.2}$$

The projective vector $[s_1 : \cdots : s_{2m}]$, and hence every propagation ratio s_i/s_j , depends only on C .

For $\lambda, \mu \in \mathbb{F}_q$, put

$$F_i^-(\lambda) = \begin{pmatrix} 1 & 0 \\ \lambda s_i & 1 \end{pmatrix}, \quad F_i^+(\mu) = \begin{pmatrix} 1 & \mu s_i^{-1} \\ 0 & 1 \end{pmatrix}. \quad (6.3)$$

On the CSS label space $L_C = (C \oplus 0) + (0 \oplus C^\perp)$, the product of the lower blocks sends

$$(c, h) \mapsto (c, h + \lambda S c),$$

and the product of the upper blocks sends

$$(c, h) \mapsto (c + \mu S^{-1} h, h).$$

Thus both products preserve L_C and lift to product-Clifford tensor symmetries: Lemma 4.1 supplies the required product-Pauli corrections. Thus label-space preservation is sufficient here; no coherent choice of Clifford phases is being assumed. At any chosen input party j , their blocks run through all lower and upper elementary unipotents because $s_j \neq 0$. Together these generate $\mathrm{SL}_2(q)$. Under the Choi correspondence, an input block A induces the logical block A^{-T} . This map is an automorphism of $\mathrm{SL}_2(q)$, so every one-qudit logical symplectic action has a product-Clifford physical implementation. Logical Paulis have physical Pauli representatives, so the full projective one-qudit Clifford group $\mathbb{F}_q^2 \rtimes \mathrm{SL}_2(q)$ is transversal.

Corollary 1.2 excludes every product-unitary logical action outside that group. Hence the displayed group is exact.

For a generalized or extended generalized Reed–Solomon code, the standard dual formula supplies (6.2), including the extended evaluation point at length $q+1$ [HP03, Sections 5.2–5.3]. Thus the GRS tower lies on the diagonally isodual branch, but the proof used no evaluation structure.

There are two different lift questions behind this projective statement. Fix an input coordinate j , put $t_i = s_i/s_j$, and write

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(q).$$

Propagation to coordinate i is

$$\phi_i(A) = \begin{pmatrix} a & t_i^{-1}b \\ t_i c & d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & t_i \end{pmatrix} A \begin{pmatrix} 1 & 0 \\ 0 & t_i^{-1} \end{pmatrix}. \quad (6.4)$$

Thus every ϕ_i is a homomorphism, ϕ_j is the identity, and the elementary matrices in (6.3) satisfy all $\mathrm{SL}_2(q)$ relations coordinatewise. This calculation also applies at the extended evaluation point. It uses only the diagonal duality $SC = C^\perp$, not the evaluation formula for a GRS code.

For the present odd prime q , and more generally for any odd prime power, the finite Stone–von Neumann representation of the Heisenberg group H_q extends to a genuine representation of $\mathrm{SL}_2(q) \ltimes H_q$ [Gér76, Theorem 1]. Apply its Weil factor to each $\phi_i(A)$. The resulting tensor product is multiplicative in A , and exact Egorov covariance preserves the CSS stabilizer line. Consequently the one-qudit Clifford scalar extension, restricted to the linear $\mathrm{SL}_2(q)$ factor, splits coherently with the propagated tensor symmetries. The exceptional uniqueness clause at $q = 3$ in Gérardin’s theorem does not remove existence.

The affine extension does not split. Choose $a, b \in \mathbb{F}_q$ with $\chi(ab) \neq 1$, and put $X = X(a)$, $Z = Z(b)$. Their projective classes commute, whereas every choice of unitary representatives satisfies

$$ZX = \chi(ab)XZ, \quad \chi(ab) \neq 1.$$

Scalar changes of X and Z do not change their commutator. Hence the translation subgroup $\mathbb{F}_q^2$ has no homomorphic lift to the one-qudit unitary Clifford group. The coherent lift above is therefore $H_q \rtimes \mathrm{SL}_2(q)$, not a section of $\mathbb{F}_q^2 \rtimes \mathrm{SL}_2(q)$ through the scalar quotient. This is the Heisenberg central extension, not a residual metaplectic or Schur-multiplier obstruction on $\mathrm{SL}_2(q)$.

Neither conclusion decides the party-permutation extension in Corollary 5.7. Its quotient is the evaluation-set-dependent realized permutation group Π , and its kernel is the full fixed-party projective group Γ , rather than the scalar phases. Its splitting remains exactly the nonabelian factor-set triviality criterion of Appendix C. $\square$

For a stabilizer state, the hypotheses of Corollary 3.3 hold whenever every party belongs to a support S of size at least three whose supported stabilizer label subgroup has order q^2 and projects isomorphically onto the local Weyl-label group at every party of S . No $\mathbb{F}_q$ -linearity of that subgroup is needed, in keeping with the arbitrary-additive scope of Theorem 1.1; and by Corollary 3.6 even the order q^2 may be relaxed, provided the labels recovered at each party generate.

7 The non-GRS pencil and its quotient

Our first six-party application combines atlas classification with the rigidity theorem to turn a classical projective quotient into a complete LU coordinate over odd prime fields. Let H_t be the ordered matrix in (1.2), over a finite field of odd order, and put

$$G(t) = t^4 - 4t^3 + 7t^2 - 4t + 1.$$

The admitted non-GRS condition is

$$t(t-1)(t^2-t+1)(t^2-3t+1)G(t) \neq 0. \quad (7.1)$$

Factoring the twenty 3-by-3 column minors gives the first four factors, which keep the arc and quotient coordinates nondegenerate. The determinant for a conic through the six columns is a unit times G , so $G = 0$ is exactly the GRS boundary. Write $C_t = \ker H_t$ and $\Psi_t = \Psi_{C_t}$, and let A, B, z be as in (1.3).

Theorem 7.1 (Projective and Clifford quotient). For admitted t, u , the following are equivalent:

$$H_t \sim_{\mathrm{proj}} H_u, \quad C_t \sim_{\mathrm{mon}} C_u, \quad z(t) = z(u),$$

allowing arbitrary party permutations. If q is prime, these are also equivalent to $\Psi_t \sim_{\mathrm{LC}} \Psi_u$.

Proof. Set

$$y(t) = \frac{(t-1)^2}{t}.$$

Then $A = -4t^2y$, $B = t^2(y^2 - 1)$, and

$$z = \frac{(y - y^{-1})^2}{16}.$$

It follows that $z(t) = z(u)$ exactly when

$$y(u) \in \{y(t), -y(t), y(t)^{-1}, -y(t)^{-1}\}. \quad (7.2)$$

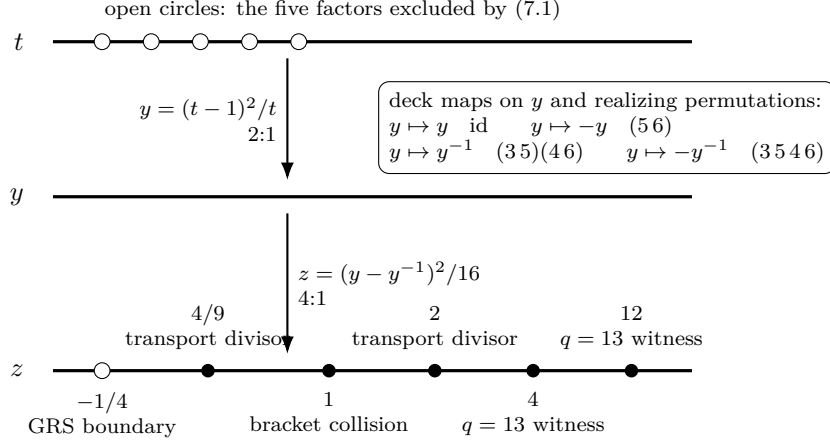


Figure 6: The degree-eight quotient of the admitted pencil (special points on the z -line are arranged schematically). The parameter t double-covers $y = (t - 1)^2/t$, and the four identifications $y \mapsto \pm y^{\pm 1}$, each realized by a displayed party permutation, cover z . The open point $z = -1/4$ is the excluded GRS boundary; $z = 1$ is a bracket collision; $z = 4, 12$ form the $q = 13$ four-copy witness pair; and $z = 2, 4/9$ belong only to the transport-rank detector, not to the LU-orbit boundary.

Each of the four relations is realized by a direct projectivity and a party permutation. In order, one may use

relation	projectivity
$y(u) = y(t)$	$\text{diag}(1, (1 - u)/(1 - t), u/t)$
$y(u) = -y(t)$	$\text{diag}(1, (1 - u)/(1 - t), -u/t)$
$y(u) = y(t)^{-1}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & (u-1)/t \\ 0 & -u/(1-t) & 0 \end{pmatrix}$
$y(u) = -y(t)^{-1}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & (1-u)/t \\ 0 & -u/(1-t) & 0 \end{pmatrix}$.

The corresponding permutations are respectively the identity, $(5\ 6)$, $(3\ 5)(4\ 6)$, and $(3\ 5\ 4\ 6)$. All denominators and determinants are nonzero on the admitted locus.

Conversely, the ten signed products of complementary 3-by-3 brackets of H_t form the multiset

$$\{+A, +A, +A, -A, -A, -A, +B, +B, -B, -B\}. \quad (7.3)$$

Projectivity and independent column rescaling multiply all ten products by one common scalar; party permutation changes at most their common sign. On the admitted locus $A, B \neq 0$. If $A^2 \neq B^2$, the multiplicities three and two in (7.3) distinguish the two opposite pairs and recover $(B/A)^2 = z$. If $A^2 = B^2$, the multiset instead collapses to two opposite values, each of multiplicity five. That collapsed pattern is itself invariant under the common scaling and sign, and it is equivalent to $z = 1$. Thus the multiset recovers z in both cases, and projective equivalence forces equality of z . The equivalence with monomial codes follows from Section 2, and projective equivalence supplies an LC equivalence.

We now specialize the transition atlas from the rigidity argument to $m = 3$. Each four-party support identifies its two-dimensional stabilizer plane with the local Pauli-label plane at any party in that support. Overlapping supports therefore give coordinate changes between local Weyl frames, and their compositions around a loop give holonomy up to local symplectic conjugacy. By (7.2), these minimum-support planes generate the full stabilizer label group. The 450 quantities below are conjugacy data of this atlas, not an independent choice of scalar contractions.

For the remaining direction assume that q is prime. Then a local Clifford induces a block $F_i \in \mathrm{SL}_2(q)$ on each local Pauli-label plane. Put $K_T = L_{C_i}(T)$ for every four-set T . If $i, j \in T$, projection $p_i : K_T \rightarrow \mathbb{F}_q^2$ onto the Pauli-label plane at i is an isomorphism by (7.4). Define the transition

$$M_T(i, j) = p_j p_i^{-1} : (\mathbb{F}_q^2)_i \longrightarrow (\mathbb{F}_q^2)_j$$

and, for distinct four-sets T, U containing i, j , the holonomy

$$\mathrm{Hol}(T, U; i, j) = M_U(j, i) M_T(i, j). \quad (7.4)$$

Local symplectic blocks F_i send this endomorphism to $F_i \mathrm{Hol} F_i^{-1}$. Reversing T, U takes its inverse, and a party permutation only relabels the data. Thus the multiset of

$$r(M) = \mathrm{Tr}(M)^2 / \det M$$

over the $15 \cdot 6 \cdot 5 = 450$ choices is an LC and party-permutation invariant.

Substituting the shortened generators into (7.4), simplifying each of the 450 rational functions exactly, and collecting equal functions gives the following multiset. The paper-local certificate described in Section 10 replays both this symbolic enumeration and a separate direct-Lagrangian reconstruction:

value of $r(M)$	multiplicity
4	90
$-1/z$	144
$(2z+1)^2/z^2$	24
each root of $X^2 - 8X + (8 - 16z - 1/z)$	96.

(7.5)

Because $A, B \neq 0$ on the admitted locus, $z \neq 0$. The two roots are taken over an algebraic closure; their symmetric multiset descends to $\mathbb{F}_q$. The multiplicity 96 applies to each root before specialization; coincident values have their multiplicities added. The identity

$$4B(t)^2 + A(t)^2 = 4G(t)^2$$

shows that the excluded GRS locus is exactly $z = -1/4$. Away from it, the constant 4 bin is isolated and the two quadratic roots are distinct. The bin containing $-1/z$ can merge with neither, one, or both of the weight-24 and one weight-96 contributions, so its multiplicity is 144, 168, 240, or 264. No other bin has one of these multiplicities. Hence the multiset canonically recovers $-1/z$, including every modular collision, and LC equivalence forces equality of z . $\square$

The map $t \mapsto z$ has degree eight: the four transformations in (7.2), followed by the quadratic relation $t+t^{-1} = y+2$, account for its eight sheets. The signed refinement $w = B/A$, $z = w^2$, is not quantum data. Both even and odd explicit party permutations reverse w , and their projectivities lift to monomial local Cliffords.

Proof of Corollary 1.5. Theorem 7.1 gives all equivalences except LU. Every LC is an LU, while Theorem 1.1 turns every LU intertwiner between the states Ψ_t, Ψ_u into an LC intertwiner. $\square$

Proposition 7.2 (Frobenius-sector divisors). Let σ be an automorphism of a finite field of odd order and let t be admitted and non-GRS. Put

$$E_\sigma(t) = (\sigma(t) - t)(1 - \sigma(t)t), \quad D_\sigma(t) = ((1 - \sigma(t))(1 - t))^2 + \sigma(t)t.$$

Then

$$\frac{\sigma(t)}{(1 - \sigma(t))^2} = \frac{t}{(1 - t)^2} \iff E_\sigma(t) = 0. \quad (7.6)$$

If $h = (1 - \sigma(t))(1 - t)$ and $w = (-h, -h, 1, 1, -1, -1)$, then

$$H_{\sigma(t)} \text{diag}(w) H_t^\top = 0 \iff D_\sigma(t) = 0. \quad (7.7)$$

Moreover $D_{\text{id}}(t) = G(t)$, and

$$E_\sigma(t) = 0 \implies D_\sigma(t) \neq 0. \quad (7.8)$$

Finally G, A, B, z commute with σ , and σ preserves the admitted non-GRS locus.

Proof. Admission gives $t \neq 1$, and injectivity of σ gives $\sigma(t) \neq 1$. Cross multiplication in (7.6) gives

$$s(1 - t)^2 - t(1 - s)^2 = (s - t)(1 - st)$$

with $s = \sigma(t)$. Direct multiplication in (7.7) makes eight entries vanish identically and leaves $-2D_\sigma(t)$ in the last entry; odd characteristic gives the equivalence. Setting σ to the identity yields $D_{\text{id}} = G$.

If $E_\sigma(t) = 0$, then either $\sigma(t) = t$ or $\sigma(t)t = 1$. In the first case $D_\sigma(t) = G(t)$. In the second, multiplying $D_\sigma(t)$ by t^2 gives $G(t)$. Admission excludes $t = 0$ and $G(t) = 0$, proving (7.8). The equivariance statements follow by applying σ to the defining polynomial and rational expressions. $\square$

The prime-field hypothesis is necessary for this scalar classification. Over $\mathbb{F}_{p^e}$, the basis permutation $|x\rangle \mapsto |x^p\rangle$ is a local Clifford and sends Ψ_t to Ψ_{t^p} , while $z(t^p) = z(t)^p$ need not equal $z(t)$. Concretely, in $\mathbb{F}_{25} = \mathbb{F}_5[s]/(s^2 - 2)$, both s and $-s = s^5$ are admitted, but

$$z(s) = 3 + 3s, \quad z(-s) = 3 + 2s.$$

For each field automorphism σ , Proposition 7.2 gives the collision and diagonal-isodual divisors relating t to $\sigma(t)$. It does not show that an arbitrary additive local Clifford determines a nontrivial field automorphism, nor does it construct a local Clifford from a Galois match of z . Those two representation-theoretic bridges are not claimed here, so neither is a full extension-field LC/LU classification.

8 Diagonal isoduality and the logical-Clifford phase

The second application uses the atlas gauges to determine the exact fixed-party transversal logical group of an MDS-CSS encoder. Throughout this section q is an odd prime. Regard any one party of an $\text{AME}(6, q)$ stabilizer tensor as an input and the other five as outputs. This is a $[[5, 1, 3]]_q$ encoding isometry. In prime local dimension, the symplectic action of a one-qudit Clifford is a block in $\text{SL}_2(q)$, so the kernel of the party action on Clifford tensor symmetries is

$$\mathcal{K}_C = \{(F_1, \dots, F_6) \in \text{SL}_2(q)^6 : (\bigoplus_i F_i) L_C = L_C\}.$$

For $q = p^e$ with $e > 1$, the full local Clifford group acts instead through $\text{Sp}_{2e}(\mathbb{F}_p)$; the displayed field-linear block group is then not the full local Clifford kernel. For an encoder view, the *fixed-party logical symplectic image* means the image of $\mathcal{K}_C$ on its input block. Corollary 1.4 identifies the all-length phase boundary as diagonal isoduality. For six-arcs, classical self-association turns that intrinsic code condition into the conic criterion below.

Lemma 8.1 (Fixed-label self-association). Let H be a rank-three 3×6 matrix whose columns are a six-arc, and put $C = \ker H$. There is a nonsingular diagonal matrix S with $SC = C^\perp$ if and only if the six columns of H lie on a conic.

Proof. Let G be a 3×6 generator matrix for C . Its columns are the labeled Gale transform of the columns of H . Since $C^\perp = \text{row}(H)$,

$$SC = C^\perp \iff \text{row}(GS) = \text{row}(H) \iff AGS = H \text{ for some } A \in \text{GL}_3(q).$$

Thus the condition is precisely fixed-label projective equivalence to the Gale transform. This fixed locus can be identified directly over the finite field. Put

$$V_i = (x_i^2, y_i^2, z_i^2, x_i y_i, x_i z_i, y_i z_i)$$

and let V be the 6-by-6 matrix with rows V_i . Its determinant vanishes exactly when a nonzero quadratic form contains all six columns. Every five rows of V are independent: a relation supported on at most five points would give $H_I D H_I^T = 0$ for a nonsingular diagonal D on its support I ; for $|I| \geq 3$, this would put the three-dimensional row space of $H_I D$ inside $\ker H_I$, which has dimension at most two, while smaller supports are immediate. Hence any relation $\sum_i d_i V_i = 0$ has every $d_i \neq 0$.

Writing $D = \text{diag}(d_i)$, that relation is $HDH^T = 0$. Therefore $\text{row}(HD) \subseteq \ker H = C$, and equality follows from dimension three. Thus $CD^{-1} = C^\perp$. Conversely, a nonsingular diagonal equivalence $SC = C^\perp$ gives $HS^{-1}H^T = 0$, hence a relation among the rows of V and a conic through the six columns. The conic is irreducible because the columns form an arc. This is the finite-field fixed-label form of the classical six-point self-association theorem [HMSV08, Section 2.3]. $\square$

Theorem 8.2 (Fixed-party logical phase). For an equal-phase CSS state from a six-arc over an odd prime field, the fixed-party logical symplectic image is

$$\begin{cases} \text{SL}_2(q), & \text{if the arc is conic/GRS,} \\ T = \{\text{diag}(a, a^{-1}) : a \in \mathbb{F}_q^\times\}, & \text{if the arc is nonconic.} \end{cases} \quad (8.1)$$

Thus every encoder view of a GRS tensor realizes the full one-qudit logical symplectic group from fixed-party symmetries. Off the GRS locus, the fixed-party logical symplectic image is the split torus. At a chosen anchor leg, any party-moving isoduality that exchanges the CSS X - and Z -spaces transports the encoder view by the nontrivial Weyl-group element, so the possible anchor blocks generate the normalizer of T together with T . This is an action on one fixed encoder only when the isoduality fixes its input leg.

Proof. Corollary 1.4 gives $\text{SL}_2(q)$ exactly on the diagonally isodual branch and T otherwise. Lemma 8.1 identifies diagonal isoduality with the conic locus. At a chosen anchor leg, a party-moving isoduality exchanges the CSS X - and Z -axes and hence represents the nontrivial Weyl-group element normalizing T . It acts on one fixed encoder only when the party motion fixes that encoder's input. $\square$

Adding logical Paulis recovers the exact fixed-party projective logical groups in Corollary 1.4.

This is an operational distinction, not a classical automorphism count. The fixed-party non-GRS group has order $q^2(q-1)$, while the GRS group has order $q^2|\text{SL}_2(q)|$.

Lemma 8.3 (Coset states and three-party syndrome charts). Let H be a rank-three parity-check matrix whose six columns form an arc, put $C = \ker H$, and set

$$|\Psi_e\rangle = X(e)|\Psi_C\rangle, \quad |\Psi_C\rangle = q^{-3/2} \sum_{c \in C} |c\rangle.$$

Then $|\Psi_e\rangle = |\Psi_f\rangle$ if and only if $e - f \in C$, and states from distinct cosets are orthogonal. In the row basis of H , the $Z(C^\perp)$ -stabilizer syndrome of $|\Psi_e\rangle$ is $He^\top$. Consequently the least support of an X -operator producing $|\Psi_e\rangle$ is the minimum Hamming weight in $e + C$.

For every syndrome $s \in \mathbb{F}_q^3$ and every three-party set I , there is a unique representative e supported in I with $He^\top = s$. If the coset with syndrome s has minimum weight three, this representative has support exactly I .

Proof. Translation replaces the computational support C by the affine coset $e + C$. Two such supports are equal precisely when $e - f \in C$; otherwise they are disjoint, which also proves orthogonality. If h is a row of H , then $h \in C^\perp$, and $Z(h)$ acts on the entire coset $e + C$ by the character $\chi(h \cdot e)$. Reading this for the three rows of H gives $He^\top$.

Now fix I . The arc condition says exactly that the three columns in I form an invertible matrix H_I . Thus $H_I e_I^\top = s$ has one solution, and extension by zero gives the unique representative supported in I . If one of its three coordinates vanished, the same syndrome would have a representative of weight at most two. A minimum weight-three syndrome therefore uses all three coordinates. $\square$

Proposition 8.4 (Clebsch extremal X -syndrome conic). Let H be a parity-check matrix whose columns are the Clebsch six-arc in $\mathbb{P}^2(\mathbb{F}_{11})$, put $C = \ker H$, and write

$$|\Psi_e\rangle = X(e)|\Psi_C\rangle, \quad |\Psi_C\rangle = 11^{-3/2} \sum_{c \in C} |c\rangle.$$

The syndromes whose cosets have minimum X -support three are the 120 nonzero vectors on twelve projective rays forming a nonsingular conic in $\mathbb{P}^2(\mathbb{F}_{11})$. Each such syndrome has exactly one weight-three representative on each of the $\binom{6}{3} = 20$ three-party supports. The monomial automorphism group $C_{10} \times A_5$ is transitive on the 120 syndromes: A_5 is transitive on the twelve rays with point stabilizer C_5 , and C_{10} is transitive on the ten nonzero vectors of each ray. Nevertheless, the defining six-arc is nonconic, so the fixed-party logical symplectic image of every encoder view is the split torus T , not $\mathrm{SL}_2(11)$.

Proof. Lemma 8.3 identifies minimum X -support with coset-leader weight and gives one representative on every three-party support. The Clebsch deep-hole theorem identifies the covering radius as three and the twelve projective maximum-distance syndromes with a nonsingular conic; it also gives the transitive $C_{10} \times A_5$ action [Rud26, Proposition 6.2, Corollary 6.3, and Proposition 6.4]. Finally, the Clebsch arc is non-GRS and hence nonconic. The last assertion is therefore Theorem 8.2 at $q = 11$. $\square$

Remark 8.5 (Error symmetry without fixed-party duality). The one-per-support statement means that the same extremal translated state can be created by a minimum X -operator on either side of every balanced $3 \mid 3$ cut. The conic belongs to the X -error syndrome plane; it does not put the six parity-check columns on a conic. Thus the extremal X -error shell has icosahedral covariance while the fixed-party transversal group has no Fourier block exchanging X and Z . This is a fixed-party statement, not an obstruction to every electric-magnetic exchange. In the computed Clebsch row of Corollary C.1, the factor C_{10} scales the ten nonzero syndrome vectors on each ray and disappears after projectivization. The remaining A_5 is the projective monomial quotient, whereas S_5 is the larger realized party image. Odd party motion inverts T and supplies the nontrivial class of $N(T)/T$.

Changing the row basis of H changes the displayed conic by a projective coordinate transformation. Moreover, the covering radius here concerns only generalized- X translates of the six-leg AME state; it is not a claim that the associated $[[5, 1, 3]]_{11}$ output code corrects arbitrary weight-three errors. The unlabelled conic also carries neither a chosen coordinate six-set nor the support orientation used by finer Clebsch constructions. Theorem 1.1 supplies the separate exact rigidity statement: every product-unitary intertwiner from this tensor to a stabilizer AME tensor is Clifford factor by factor.

A different MDS uniformity governs operator pushing. Fix an input party and a nonidentity Pauli label. Each of the ten four-sets containing that party carries a unique stabilizer representative with that support; removing the input leaves a weight-three output Pauli. Thus a fixed input Pauli has ten minimum-support pushes. Lemma 8.3 instead fixes an X -syndrome and varies over all twenty three-party supports. Neither count uses the finite group calculation, although the symmetry orbits of the operator pushes can distinguish finer projective classes.

9 Scalar invariants and generic blindness

The operator-valued marginal mechanism retains local Weyl axes. Fixed-copy scalar contractions retain less: on algebraic families they are generically constant and can distinguish parameters only on rank-jump strata. We make that limitation precise here; Appendix A records two explicit separators on exceptional strata.

For r ket copies and r bra copies, choose a permutation $\sigma_i \in S_r$ at each party and contract ket copy a with bra copy $\sigma_i(a)$. If G generates a d -dimensional code, let $M_{\sigma}(G)$ be the homogeneous matching system

$$(u_a G)_i = (v_{\sigma_i(a)} G)_i \quad (1 \leq i \leq n, 1 \leq a \leq r).$$

Counting its solutions gives the normalized contraction

$$I_{\sigma}(\Psi_C) = q^{dr - \text{rank } M_{\sigma}(G)}. \quad (9.1)$$

Proposition 9.1 (Generic constancy). Fix r and a field k of cardinality q , which is also the local Hilbert-space dimension under the code-state dictionary. Fix an irreducible reduced parameter variety X over k , and a regular generator-matrix chart $G(x)$ of constant code dimension. Every degree- (r, r) permutation contraction of the associated equal-phase states has one value at every k -point of some dense open subscheme $U \subseteq X$ defined over k . This asserts no existence of k -points in U : if $U(k) = \emptyset$, the finite-field pointwise conclusion is vacuous. The same holds for party-orbit sums and linear combinations. After evaluating the states as vectors in $(\mathbb{C}^q)^{\otimes n}$, in the stable range $q \geq r$, every scalar LU-invariant of bidegree (r, r) is generically constant in this sense. For a reducible family, the statement holds separately on each irreducible component.

Proof. Formula (9.1) depends only on the rank of a matrix whose entries are regular functions over k . A maximal nonzero minor shows that this rank is constant on a dense open subscheme defined over k . There are only finitely many diagrams at fixed r , so their generic-rank opens have a common dense intersection because X is irreducible. This proves the geometric generic-rank assertion; the explicit caveat in the statement is necessary when passing to finite-field rational points. Over the complex vector space $(\mathbb{C}^q)^{\otimes n}$, applied pointwise to each code state, the first fundamental theorem for unitary invariants says that local copy-permutation diagrams span the bidegree- (r, r) invariants; when the local Hilbert-space dimension q is at least r , there are no additional dimension relations among these diagrams [CS06, Section 2]. $\square$

Thus no fixed copy number can recover a nonconstant pencil coordinate on the generic locus. A scalar witness can still detect a special rank-jump stratum, as the four-copy example in Appendix A does. In contrast, the four-party operator tensor (3.2) retains the Weyl axes and forces Cliffordness. This is the precise distinction between scalar blindness and marginal covariant rigidity.

10 Verification and scope

The rigidity theorem, the arc-AME dictionary, and generic constancy are conceptual proofs in the text. The projective quotient uses displayed bracket identities and projectivities. Its transition-map invariance and collision argument are proved in Section 4, while the 450 exact rational functions are certificate-supported. Section 6 proves the general MDS diagonal-multiplier criterion. Section 8 combines it with Gale duality to obtain the six-arc phase theorem. The party groups, factor sets, complements, and torus actions in Corollary C.1 are certificate-supported finite examples, not inputs to the uniform theorem. Appendix A proves the marginal trace/rank and incidence bridges; the H_3 determinants and $q = 13$ contractions are certificate-supported. Appendix B proves the systematic transport reduction; the cycle-cover determinants, assignment exhaustion, exact minors, and double-coset support sets are certificate-supported.

The reproducibility package is in `supplement/`. From that directory,

```
python3 verify.py          checks all seventeen byte counts and SHA-256 digests,
python3 verify.py --replay regenerates all eight canonical JSON certificates in memory.
```

The full replay uses only the Python standard library, deterministic integer, rational, polynomial, and finite-field arithmetic, and canonical JSON serialization. It includes separately implemented checks appropriate to each claim: CSS versus direct-Lagrangian shortening, symbolic versus finite projectivity, chord concurrency versus stabilizer ranks, quotient minors versus finite-field elimination, and cycle-cover versus fraction-free transport determinants. These checks belong to one paper-owned Python package. They provide independent implementations within that package, not an independent external reproduction. The exact files, entry points, hashes, domains, expected outputs, and negative scopes are recorded in `supplement/EVIDENCE.md` and `supplement/EVIDENCE-MANIFEST.json`. On the reference machine the digest check takes under a second, the full evidence replay about six minutes, and each Lean aggregate under ten seconds.

From the repository root, the exact formal replay is

```
python3 lean/scripts/lean-build-queue.py run \
  RelativeConicArcs.Gates.AMELUAggregate \
  RelativeConicArcs.Gates.AMELUAggregateAxioms \
  --profile single --threads 1 \
  --aggregate RelativeConicArcs.Gates.AMELUAggregateAxioms
```

The checked release is then reproduced from `papers/ame_lu/` by

```
make release-check
```

The release identifier is content based: `release/RELEASE-MANIFEST.json` records the SHA-256 digest of every public byte and every file in the recursive formal dependency closure, including the Lean toolchain and build definitions. The manifest, rather than a mutable working-tree name, is the immutable identity of the companion release.

Claim-to-trust map. LU-to-LC rigidity, atlas classification, and the transversal no-go. The manuscript gives complete proofs. Lean kernel-checks the support, transition, marginal, covariance, axis-recovery, and holonomy cores; the manuscript proves the exact projective atlas extension.

Partial-Weyl recognition criterion. Section 3.1 gives complete manuscript proofs of the partial-marginal criterion, the recognition-group lemma, its minimal-support realization, and the CSS and arbitrary-dimension consequences. These results use no finite certificate and have no Lean coverage. Their relation to the qubit theorem of Van den Nest, Dehaene, and De Moor and to the tensor-uniqueness literature is established by the cited sources and the manuscript reduction, not by the supplement.

Discreteness and quantitative stability. The manuscript gives complete proofs. No certificate and no Lean module supports Section 5; the numerical checks that first accompanied these statements are not used and are not part of the supplement. Every constant in the subsection is exhibited. The threshold in Corollary 5.19 is obtained by three explicit routes with different standing: Corollary 5.19 produces it from compactness without exhibiting it. Theorem 5.29 gives the single-marginal closed formula, while Theorem 5.25 aggregates all orthogonal minimum-support sectors and gives the stronger low-dimensional frame radius. The latter formula is polynomial in the party count at local dimension two, conditional on existence, with no claim of optimality. Theorem 5.28 bypasses support amplitudes by a three-region correctable-code commutator and gives the strongest radius, of order $n^{-1/2}$ for every fixed local dimension, again with no optimality claim. Its exact qualitative mechanism is inherited from the cleaning and transversal-gate hierarchy literature [BT09, BK13, PY15, JOKY18]; the manuscript-local claims are the leakage-aware 8ε commutator estimate, the Weyl-Fourier rounding constant, and their AME global composition. Proposition 5.15 bounds any threshold valid at uniformity order three; it is a statement about that order, not about the party count, and the certified region is measured against the uniformity order throughout.

Pencil and six-arc classification. The manuscript proof uses exact certificates for the finite holonomy calculation. The Lean interface is conditional and names its geometric and holonomy inputs.

H_3 and $q = 13$ separators. The manuscript proof uses exact certificates for the determinants and contraction ranks. The finite graph core is unconditional in Lean; the trace/covariance bridges and exact ranks are named inputs.

Diagonal-isodual logical groups. The manuscript gives the proof, and the multiplier theorem is unconditional in Lean. The formal exact-carrier terminal names the propagation interface as a hypothesis.

Clebsch extremal X -syndromes. The coset-state, stabilizer-character, and one-representative-per-support lemma is proved in the manuscript and kernel checked in Lean. The nonsingular conic, 120 syndromes, and transitive $C_{10} \times A_5$ action are imported from the cited companion Clebsch theorem; they are not outputs of this paper's certificates or Lean aggregate.

Twelve computed party-extension splittings and transport. Exact certificates supply the finite complements and transport matrices. Lean proves the abstract splitting algebra once a complement is supplied.

The corresponding certificate entry points and formal theorem names appear in the claim crosswalk in `supplement/EVIDENCE.md`.

The trusted boundary is Python integer arithmetic, the exact-arithmetic implementations in the generators, deterministic Gaussian elimination, and the mathematical dictionaries connecting arcs, codes, stabilizers, and tensor contractions. The certificates check the bounded enumerations and symbolic identities that they record. They do not prove the conceptual correspondence between the implemented objects and the paper definitions; that correspondence is supplied in the preceding sections.

The Lean 4 artifact uses toolchain `v4.32.0-rc1`; no aggregate module uses `sorry` or an unsafe declaration. Its aggregate contains unconditional formal proofs of the support squeeze, minimum-support generation, marginal and axis-recovery cores, atlas equivalence, abstract holonomy centralizers, the Choi/transversal bridge, the diagonal-multiplier theorem, and the abstract projective-extension algebra. The ordinary kernel-checked declarations use only `propext`, `Classical.choice`, and `Quot.sound`; three finite graph cardinalities use `native_decide` and are reported separately by the axiom terminal. All other finite evidence is replayed by the Python supplement rather than imported as a Lean axiom.

The geometric classification, exact logical propagation, finite contractions, concrete party complements, and transport-orbit inputs enter conditional interfaces. The manuscript proves or the supplement certificate-checks those inputs as recorded in the claim-to-trust map above; importing the aggregate does not turn them into unconditional Lean theorems. The exact module-to-claim crosswalk and each named hypothesis are recorded in `supplement/EVIDENCE.md`.

The extension-field scalar statements in Proposition 7.2 are unconditional in Lean. The aggregate audit checks the frame-ratio factorization, explicit Gale multiplier and odd-characteristic zero criterion, same-exponent sector disjointness, and automorphism equivariance of G, A, B, z and the admitted locus. Its separate Galois- z orbit terminal is conditional on two named bridges: extraction of a projective Frobenius sector from an additive Clifford equivalence, and construction of a Clifford from a Galois match. Neither bridge is used as a paper theorem.

Finally, none of the results classifies nonlinear orthogonal arrays, arbitrary nonstabilizer AME tensors, or stabilizer states that are not AME. Theorem 5.3 and Theorem 5.8 apply to all of them, but they bound symmetry groups rather than classify states, and they do not identify the finite group; Remark 5.6 exhibits a 2-uniform state whose symmetry group contains a non-Clifford element. We do not claim that four copies are globally minimal, that $z = 2, 4/9$ are orbit exceptions, or that a fixed copy degree separates all classes as q varies. The logical-group theorem and the quantum part of the pencil classification are restricted to odd prime fields. Over extension fields, Frobenius basis permutations are additional local Cliffords. The scalar z is Frobenius-covariant, and Proposition 7.2 determines the two divisors attached to each field automorphism, but the two additive-Clifford bridges needed for a full orbit classification are outside the theorem package. The minimum-support atlas theorem remains exact over extension fields, with local blocks in $\mathrm{Sp}_{2e}(\mathbb{F}_p)$; its holonomy form requires both a centralizing base block and the propagated symplectic membership conditions, and does not imply a semilinear classification. Theorem A.2 is restricted to the stated odd residue-field reductions, and the detector exceptions described in Appendix B do not alter the characteristic-uniform rigidity theorem.

11 Conclusion

An $(m + 1)$ -party marginal of a stabilizer $\mathrm{AME}(2m, q)$ state retains the full configuration of local Weyl axes. Rank-one contractions recover those axes intrinsically, so every product-unitary intertwiner is Clifford factor by factor. The minimum-support atlas is the structural completion of this rigidity mechanism: it records how the recovered frames fit together and classifies the residual Clifford equivalence exactly.

Two-party marginals already decide the coarse question. They make the symmetry group finite and control how fast the defect grows away from it, at any local dimension and with no stabilizer hypothesis, so the maximal entanglement and stabilizer structure used above are spent entirely on naming the finite group rather than on producing it. Naming it also separates it: stabilizer overlaps are quantized by a subgroup index, so the exact product-Clifford

symmetries stand apart in defect by a constant of the characteristic, and an approximate symmetry small enough to be rounded to a Clifford is thereby exact.

The applications expose two consequences of that structure. For the equal-phase linear MDS–CSS states over odd prime fields considered here, diagonal isoduality is exactly the boundary between the full projective logical $\mathbb{F}_q^2 \rtimes \mathrm{SL}_2(q)$ group and the affine split-torus subgroup. On the six-party pencil over odd prime fields, the degree-eight projective quotient z is also the complete LU coordinate. Fixed-copy scalar contractions, by contrast, discard the recovered axes and are generically constant on irreducible code families.

Over extension fields, the atlas classification remains exact with local blocks in $\mathrm{Sp}_{2e}(\mathbb{F}_p)$, but the two representation-theoretic bridges from additive Clifford maps to Frobenius sectors and back are not proved here. Determining when an extension-field atlas admits a semilinear canonical form is the remaining exact classification problem.

A Explicit scalar certificates

This appendix records two explicit scalar LU separators. They are not used in the uniform rigidity, atlas, or logical-group results; they show what finite-copy scalar contractions can still detect on exceptional strata despite the generic blindness proved in Section 9.

A.1 A marginal-moment separator

For a four-party set T , let ρ_T be the reduced state and embed it into all six parties as

$$A_T = \rho_T \otimes I_{T^c}.$$

For an unordered triple $\{T, U, V\}$ of distinct four-sets, the number $\mathrm{Tr}(A_T A_U A_V)$ is invariant under arbitrary product unitaries; party permutation merely permutes the 455 triples. By (2.3),

$$\mathrm{Tr}(A_T A_U A_V) = q^{-\mathrm{rank}(L_C(T) + L_C(U) + L_C(V))}. \quad (\text{A.1})$$

Here and below, ranks of label spaces and matching systems are over $\mathbb{F}_q$. Indeed, the three stabilizer sums have total normalization q^{-12} . Only triples of labels with sum zero survive the full trace; isotropy of L_C removes multiplication phases, and the kernel of the addition map from the three two-planes has size q^{6-r} , where r is the $\mathbb{F}_q$ -rank in (A.1).

Index a four-set by its omitted pair, an edge of K_6 . The rank in (A.1) is four exactly when the three corresponding chord lines concur in both the six-arc $\mathcal{A}$ and its Gale dual $\mathcal{A}^*$, represented by any kernel basis of the arc matrix. To see this, split each shortened CSS plane into its one-dimensional $X(C)$ and $Z(C^\perp)$ parts. For three omitted edges, the three shortened X -lines span dimension two precisely when the corresponding chords of $\mathcal{A}^*$ concur; the Z -lines give the same criterion for $\mathcal{A}$. Each span has dimension at least two, so their sum has rank four exactly when both concurrences hold.

Sixty triples are stars and concur universally. Any other three-edge graph that contains two adjacent edges but is not a star has two chords meeting at their common arc point and a third chord missing it, since the columns form an arc. The only remaining graph type is a perfect matching. Hence

$$\#\{q^{-4} \text{ moments}\} = 60 + b(\mathcal{A}, \mathcal{A}^*). \quad (\text{A.2})$$

where b counts perfect matchings concurrent in both configurations.

Lemma A.1 (Concurrent matchings on a conic). Let S be six distinct points on a nonsingular conic over a finite field k of odd characteristic other than 5. At most six perfect matchings of S have concurrent joining chords.

Proof. Identify the conic with $\mathbb{P}^1$. Projection from a point off the conic pairs the points in every fiber and hence induces a projective involution; conversely, the chords joining the three two-element orbits of a projective involution meet at its center. Thus the concurrent matchings are exactly the fixed-point-free involutions in

$$H = \text{Stab}_{\text{PGL}_2(k)}(S).$$

The correspondence is injective because an element of $\text{PGL}_2(k)$ is determined by its action on three points. Moreover, H acts freely on ordered triples of distinct points of S , so $|H| \mid 6 \cdot 5 \cdot 4 = 120$.

Dickson's classification, including the p -regular cases [Dic01, §§239–261], and Faber's p -irregular classification [Fab23, Theorem 1.1] now leave the following possibilities. The last column bounds involutions having no fixed point on this six-set:

H	fixed-point-free involutions
cyclic	≤ 1
dihedral	≤ 6
A_4	≤ 3
S_4	≤ 6
A_5 in its faithful degree-six action	0
C_3 or S_3 (char $k = 3$)	≤ 3 .

For the A_5 row, each involution fixes two of the six points. In the dihedral row, if the rotation order exceeded six, every orbit away from the two rotation-fixed points would have size at least that order, so no invariant six-set could exist; for rotation order at most six the two reflection classes together contribute at most six fixed-point-free elements (and the possible central half-turn contributes only when one reflection class has fixed points). In the characteristic-three semi-elementary case, the normal 3-group has order 3, while its cyclic complement embeds in $\text{Aut}(C_3)$, giving only C_3 or S_3 . The subfield groups whose order divides 120 are $\text{PSL}_2(3) \cong A_4$ and $\text{PGL}_2(3) \cong S_4$, already listed. The remaining characteristic-five subfield possibilities are excluded by the hypothesis. Every row is therefore bounded by six. $\square$

Theorem A.2 (Uniform H_3 /GRS separation). Let $\tau^2 = \tau + 1$, and let $\mathfrak{p}$ be a prime ideal of $\mathbb{Z}[\tau]$ of odd residue characteristic for which the reduction of the integral H_3 (icosahedral) sextuple

$$((0, 1, 1 - \tau), (0, 1, \tau - 1), (1, 1 - \tau, 0), (1, \tau - 1, 0), (1, 0, -\tau), (1, 0, \tau))$$

is a six-arc. Write $k = \mathbb{Z}[\tau]/\mathfrak{p}$. If the reduced arc is non-GRS, put $q = |k|$. Its equal-phase state is not LU-equivalent, even after party permutation, to any six-point GRS state over k . (The hypothesis is vacuous in characteristic five, where the reduction is itself GRS.)

Proof. Exact integral chord determinants give ten common concurrent matchings for the H_3 arc and its Gale dual. The other five determinant norms are -64 and -4 , so the count is exactly ten in every residue reduction covered by the theorem. This is an exact fifteen-matching calculation in $\mathbb{Z}[\tau]$, not a field sample. The H_3 moment multiset therefore contains 70 copies of q^{-4} .

Lemma A.1 gives at most six concurrent matchings on a conic. The bound is attained whenever $\text{PGL}_2(q)$ contains an octahedral S_4 , by the corresponding six-set: the six half-turns about axes through pairs of opposite edges act without fixed vertices. Thus a GRS state has at most 66 copies of q^{-4} , and the margin $70 > 66$ is sharp for this argument. In characteristic five no subgroup bound is needed: $X^2 - X - 1 = (X - 3)^2$, so the residue of τ is 3, and direct substitution gives $G(\tau) = 0$. The H_3 reduction is therefore GRS, contrary to the theorem's non-GRS hypothesis. Equation (A.1) is LU-invariant, so the gap proves the claim. $\square$

The five nonconcurrent H_3 matchings form a one-factorization of K_6 . The marginal incidence recovers its S_5 stabilizer but forgets the even A_5 orientation. This is necessary: full local Fourier transform sends Ψ_C to $\Psi_{C^\perp}$, and an integral isoduality returns the H_3 code through an odd pentad permutation in every odd reduction. The orientation is an LC-forgettable CSS electric/magnetic marking, not a separate LU class.

A.2 The exact dimension-thirteen four-copy witness

Use the permutation contraction I_σ and matching system $M_\sigma(G)$ from (9.1).

Theorem A.3 (Exact four-copy separator at $q = 13$). The two admitted $q = 13$ pencil classes with $z = 4$ and $z = 12$ are not local-unitary equivalent, even after arbitrary party permutation.

Proof. At $q = 13$, the admitted pencil has two classes in the same complete three-copy and marginal-moment bucket, with $z = 4$ and $z = 12$. Use the explicit four-copy pattern

$$\sigma = (1, (03)(12), (23), (02)(13), (01)(23), (021)).$$

For $\pi \in S_6$, put $(\pi\sigma)_i = \sigma_{\pi^{-1}(i)}$, and sum (9.1) over all party permutations:

$$J_\sigma(\Psi) = \sum_{\pi \in S_6} I_{\pi\sigma}(\Psi).$$

Over $\mathbb{F}_{13}$, the $z = 4$ class has rank 21 for all 720 terms. The $z = 12$ class has rank 20 for 192 terms and rank 21 for 528. Therefore

$$J_\sigma(z = 4) = 720 \cdot 13^{-9}, \quad J_\sigma(z = 12) = 3024 \cdot 13^{-9}. \quad (\text{A.3})$$

This proves arbitrary-LU inequivalence directly. The complete two- and three-copy contraction sets do not separate this pair; (A.3) makes no claim that four copies are globally minimal. $\square$

Over $\mathbb{Q}(t)$, the matching matrices of every fixed four-copy diagram have generically constant rank. Consequently, after finite-field specialization on the common generic-rank open, their contraction values are constant and do not recover z generically. The witness (A.3) detects a rank-jump stratum. In contrast, the four-party operator tensor (3.2) retains the Weyl axes and forces Cliffordness. This is the precise distinction between scalar blindness and marginal covariant rigidity.

B Transport calculation for the four-copy witness

We now explain the rank jump behind (A.3). Fix the six copy permutations

$$\sigma = (1, (03)(12), (23), (02)(13), (01)(23), (021)).$$

Let V be the three-dimensional message space and let $g_i : V \rightarrow k$ be the six coordinate covectors. Form a bipartite four-sheeted cover with vertices L_a, R_b , $0 \leq a, b < 4$, and a color- i edge from L_a to $R_{\sigma_i(a)}$. Put V at each vertex and k at each edge, using $g_{p(i)}$ as both restriction maps for a party assignment $p \in S_6$. A global section is a tuple $(x_a, y_b) \in V^8$ satisfying

$$g_{p(i)}(x_a - y_{\sigma_i(a)}) = 0. \quad (\text{B.1})$$

on all 24 edges. The three-dimensional constant-section space is always present. Gauging it away leaves the 24-by-21 matching matrix in (9.1).

Let $U = k^4/k(1, 1, 1, 1)$, and let ρ_i be the action of σ_i on U . For each p , the MDS property makes $g_{p(3)}, g_{p(4)}, g_{p(5)}$ a basis of V^* . Define Q_p by

$$g_{p(i)} = \sum_{k=0}^2 (Q_p)_{ik} g_{p(3+k)} \quad (0 \leq i < 3). \quad (\text{B.2})$$

Thus Q_p is the left block obtained by systematizing the party-reordered generator matrix on its last three columns; no pivot can vanish because every three coordinate covectors are independent. Put $\tilde{Q}_p = 2(t-1)Q_p$. This scalar is a unit on the admitted locus. For the identity assignment, the polynomial representative is

$$\tilde{Q}_1 = \begin{pmatrix} a & b & c \\ a & c & b \\ d & d & d \end{pmatrix}, \quad \begin{aligned} a &= -2(t-1)^2, & b &= -(t^2 - t + 1), \\ c &= -(t^2 - 3t + 1), & d &= -2(t-1). \end{aligned}$$

If $Y_k \in U$ are the three free copy vectors, then (B.1) reduces to the 9-by-9 block transport operator

$$\tilde{T}_p = ((\tilde{Q}_p)_{ik}(\rho_i - \rho_{3+k}))_{i,k=0}^2. \quad (\text{B.3})$$

The dimension identity

$$21 - \text{rank } M_{\pi\sigma}(G(t)) = 9 - \text{rank } \tilde{T}_p \quad (\text{B.4})$$

holds over the field k , where $p = \pi$ in the convention of Section 9. It follows by quotienting the three-dimensional constant-section kernel and then eliminating the three free systematic coordinates. Multiplying the resulting operator by the unit $2(t-1)$ does not change its rank. All steps are invertible on the admitted six-arc locus.

Theorem B.1 (Transport divisor). For a parameter on the admitted odd non-GRS pencil, there exists a party assignment $p \in S_6$ for which the sheaf (B.1) has a nonconstant section if and only if

$$(B^2 - 2A^2)(9B^2 - 4A^2) = 0,$$

or equivalently

$$(z-2)(9z-4) = 0. \quad (\text{B.5})$$

At a generic point above $z = 2$, exactly 96 party assignments have rank 20; at a generic point above $z = 4/9$, exactly 192 do. Every other assignment has rank 21.

Proof. For the identity systematic chart, the coefficients satisfy

$$a = b + c, \quad d^2 = -2a, \quad A = a(c - b), \quad B = bc.$$

Common copy relabelling, a change of basis in V , and rescaling a coordinate covector act on $\tilde{T}_p$ by invertible row and column operations. Modulo these operations, the 720 party assignments split into boundary types and three nonboundary relative-cover types. Here the type records the six relative copy permutations $\rho_j^{-1}\rho_i$, equivalently the even-cycle pattern in each pair of colored matchings. Expanding a representative determinant by signed cycle covers gives

$$\begin{aligned} P_- &= AB(3B - 2A), \\ P_+ &= AB(3B + 2A), \\ P_2 &= A(B^2 - 2A^2). \end{aligned}$$

With integral bases the determinants are respectively

$$64d^3 AB(3B - 2A), \quad 64d^3 AB(3B + 2A), \quad 64d^3 A(B^2 - 2A^2).$$

All omitted factors are units on the admitted odd pencil. Every assignment outside these three types has a determinant that is a unit there. The signed cycle-cover ledger, the exhaustion of the 720 assignments, and a fraction-free determinant path are recorded in the paper-local certificate. The two signed types lie above $w = B/A = \pm 2/3$, hence $z = 4/9$; the axial type lies above $z = 2$. Fraction-free elimination also gives a nonzero 8-minor on each component, so the generic kernel is exactly one-dimensional.

The row/column equivalences organize the exceptional assignments into double cosets. The axial stabilizers have orders 48 and 16, with intersection of order 8, so its support has size

$$48 \cdot 16/8 = 96.$$

For either signed type the corresponding orders are 24, 48, and 6, giving

$$24 \cdot 48/6 = 192.$$

The two signed types are distinguished by the sign of w , while the axial cycle pattern differs from both; hence their support sets are disjoint. Explicit representatives, stabilizer intersections, and complete support sets are in the certificate. Equation (B.4) turns the one-dimensional transport kernels into rank 20 of the matching matrix. $\square$

The exceptional arithmetic belongs to this detector, not to Theorem 1.1. The identity $4B^2 + A^2 = 4G^2$ gives, respectively,

char k	specialized identities
3	$B^2 - 2A^2 = G^2, \quad 9B^2 - 4A^2 = -A^2,$
5	$9B^2 - 4A^2 = 4G^2,$
7	$9B^2 - 4A^2 = 2(B^2 - 2A^2).$

Thus in characteristic 3 both admitted components disappear; in characteristic 5 the $z = 4/9$ component lies on the excluded GRS boundary; and in characteristic 7 the two components merge. The disjoint 96- and 192-element supports then give the exact rank histogram: 288 assignments have rank 20, and 432 have rank 21. Characteristics 11, 13, 41 introduce only pullback ramification, with no further rank drop. These reductions and the nonzero 8-minors that control them are replayed in the certificate.

The divisor records where one scalar contraction changes rank. It is not an LU-orbit boundary and, by Proposition 9.1, cannot be promoted to a generic coordinate.

C Computed party-permutation extensions

This optional appendix records the splitting obstruction for the projective party-permutation extension of Corollary 5.7, together with finite examples of it. It may be skipped without affecting the uniform rigidity, atlas, or transversal-group theorems.

The nonabelian factor set

Keep the notation of Corollary 5.7: Γ is the fixed-party projective product-unitary symmetry group, $\tilde{\Gamma}$ allows party motion, and Π is the realized permutation group, so that $1 \rightarrow \Gamma \rightarrow \tilde{\Gamma} \rightarrow \Pi \rightarrow 1$ is exact.

The extension has a section-free outer action $\Pi \rightarrow \text{Out}(\Gamma)$: conjugation by any lift of $\pi \in \Pi$ gives an automorphism of Γ , and changing the lift changes that automorphism by an inner automorphism, because elements of the kernel act by inner automorphisms. For a normalized set-theoretic section $s(1) = 1$, write

$$s(\pi)s(\rho) = f_s(\pi, \rho)s(\pi\rho), \quad f_s(\pi, \rho) \in \Gamma.$$

Then $f_s(1, \pi) = f_s(\pi, 1) = 1$, and, with α_π denoting conjugation by $s(\pi)$, associativity of three chosen lifts gives

$$f_s(\pi, \rho) f_s(\pi \rho, v) = \alpha_\pi(f_s(\rho, v)) f_s(\pi, \rho v).$$

If another normalized section is $s'(\pi) = b(\pi)s(\pi)$ with $b(1) = 1$, then substituting and commuting no kernel factors gives

$$f_{s'}(\pi, \rho) = b(\pi)\alpha_\pi(b(\rho)) f_s(\pi, \rho)b(\pi\rho)^{-1}.$$

The order of the factors is essential because Γ need not be abelian. The factor set is identically one precisely when the section preserves multiplication, so the extension splits exactly when f_s can be changed to the constant identity factor set by such a normalized b . That criterion is what the computed rows below verify.

Computed rows

Corollary C.1 (Computed party-extension splittings). The realized projective party-permutation extension splits for both non-GRS $q = 11$ pencil classes, the $q = 11$ GRS evaluation-set representatives

$$\{0, 1, 2, 3, 4, 5\}, \{0, 1, 2, 3, 4, 6\}, \{0, 1, 2, 3, 5, 6\}, \{0, 1, 2, 3, 5, 9\},$$

the enhanced-symmetry $t = -1$ rows over $\mathbb{F}_{17}$ and $\mathbb{F}_{31}$, and the H_3 representatives $(q, \tau) = (11, 4), (19, 5), (29, 6), (31, 13)$. For every listed non-GRS or H_3 row, party parity acts on the fixed split torus T by inversion. Thus, after choosing an anchor leg and allowing party motion to transport the corresponding encoder view, a realized odd party permutation enlarges the possible local linear carrier exactly from T to

$$N(T) = \langle T, J \rangle, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad J^2 = -I, \quad J \operatorname{diag}(a, a^{-1}) J^{-1} = \operatorname{diag}(a^{-1}, a).$$

Hence $N(T)/T \cong C_2$. The linear extension does not split in odd characteristic, although its projectivization does. If the party permutation fixes the input leg of a chosen encoder, this carrier acts as an actual logical enlargement; otherwise it compares two encoder views of the same six-leg tensor.

Proof. For each row, an exact finite computation reconstructs the fifteen four-support shortened planes, exhausts all 720 party permutations, and checks the propagated local symplectic blocks against the full CSS Lagrangian. After quotienting by the fixed-party group identified in Theorem 8.2, it is enough to test anchor block I on the GRS locus and anchor blocks I, J off it; this discards no candidate. In the order in which the six $q = 11$ rows are listed above, their party images are

$$S_5, \quad S_4, \quad V_4, \quad C_5, \quad S_3, \quad D_{12},$$

where D_{12} has order 12. The $q = 17$ and $q = 31$ enhanced rows have party images S_4 and S_5 , respectively, and each listed H_3 image is S_5 .

The certificate records the complete normalized factor set for each section. It also closes a complement of order $|\Pi|$ whose projection to Π is bijective, and verifies that the associated normalized cochain changes the factor set to the identity. Hence every listed extension splits. Conjugating the fixed torus by the stored lifts gives the sign action; any odd lift therefore induces the nontrivial class of $N(T)/T$. The complement is an involution in the projective party-extension group; its determinant-one anchor block is J , whose square is $-I$. $\square$

At $q = 11$, the $t = 2, z = 1$ Clebsch/ H_3 class has party image S_5 , while the $t = 10, z = 9$ non-GRS class has party image S_4 . Both extensions split, and odd party motion inverts T . After choosing an anchor leg, the possible one-qudit affine carriers across transported encoder views therefore form $\mathbb{F}_q^2 \rtimes N(T)$. This is the logical group of a fixed encoder only for the subgroup fixing its input leg; general party motion compares different encoder views. The GRS fixed-party logical group and this anchored non-GRS carrier have orders 159720 and 2420, respectively.

Arithmetic symmetry jumps do not change the logical phase. The characteristic-17 tetrahedral specialization lies on the GRS boundary and retains the full $SL_2(17)$ fixed-party image; its party image is S_4 , and the computed extension splits. The characteristic-31 A_5 specialization is non-GRS, so its fixed-party image is the split torus; its S_5 party image also splits, and odd motion enlarges the transported anchor-block carrier to $N(T)$.

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